Steering Affect in Large Language Model Agents

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ABSTRACT

Affective competence is becoming a critical capability for conversational AI, as success increasingly depends not only on what an agent says but also on how it says it. While frontier language models acquire this implicitly through large-scale post-training, smaller open-weight models suitable for private, low-cost deployment lack comparable capabilities. Existing approaches, such as prompting and fine-tuning, influence behaviour externally without directly controlling the model's internal representations. This research investigates whether affect can instead be controlled within a frozen language model through activation-level interventions.

Our preliminary work demonstrates the feasibility of this idea by formulating affective steering as a reinforcement learning problem. At each conversational turn, a controller selects an intervention in a Pleasure–Arousal–Dominance (PAD) latent space and applies it to the model's hidden activations during multi-turn persuasion. This adaptive internal control substantially improves persuasive success over unsteered baselines and prompt-based strategy planning, demonstrating that affect can function as a sequential control policy rather than a one-off modification.

Building on these promising results, we will investigate whether affective control can be made substantially more expressive and effective than is possible with a single fixed steering vector. We hypothesise that affect is distributed across multiple layers of a language model and that decomposing control into specialised affective experts operating at different depths will enable richer, more precise, and more interpretable modulation while revealing the causal roles of internal affective representations. We will further investigate whether these same latent representations can be leveraged to model the conversational partner, including personality, preferences, emotional state, and behavioural tendencies, to enable agents to personalise their affective strategies, resulting in more effective and natural long-term conversations.

This research aims to establish affective competence as a controllable, interpretable, and transferable capability of efficient open language-model agents, providing a principled alternative to relying solely on increasingly large proprietary models.

1 Introduction

Artificial intelligence systems are increasingly moving from passive tools to interactive agents. Instead of only classifying data, retrieving information, or generating isolated outputs, modern systems are now built to participate in extended interactions with people and other agents. Large Language Models are central to this shift as they can generate fluent language, maintain conversational context, use tools, simulate roles, and adapt their responses over multiple turns (Xi et al., 2025; Wang et al., 2024). As a result, they are now being developed not only for question answering, but also as assistants and tutors (Kestin et al., 2025), negotiators (Deng et al., 2024; Kwon et al., 2025), research agents (Gottweis et al., 2025; Yamada et al., 2025), and simulated participants in social environments (Park et al., 2023; Piao et al., 2025).

This shift changes what it means to control an AI system. For a single-output model, it may be enough to just check if the answer is correct, relevant or safe. But for an interactive agent, the problem is broader. The agent must now decide how to respond, when to change direction, how to strongly pursue a goal, and how to maintain the user's engagement over time. The tone, confidence, warmth, urgency, and assertiveness of a response can affect how the interlocutor interprets the agent's intent and whether the interaction continues productively.

This makes affective control an important part of LLM agent design. A dialogue agent does not only need to know what to say; it also needs some way of regulating how it says it (Picard, 1997). The same factual content can function as reassurance, encouragement, pressure, or dismissal depending on its affective stance. This is particularly important in multi-turn tasks such as tutoring, negotiation, persuasion, support, and social simulation (Deng et al., 2024; Wang et al., 2019; Liu et al., 2021; Park et al., 2023), where the appropriate stance may need to change as the interaction unfolds.

Imagine a person asking an AI assistant: "My father has chest pain, but he says it is probably just indigestion. Should I wait and see?" A purely functional system might answer by taking a position like: "No. You should not wait. Chest pain may be serious. Call emergency services". This is medically sensible, but is also abrupt. The assistant has disagreed with the user, but has not handled the user's hesitation, fear of overreacting and doesn't show concern about the family member. In practice, the user might continue negotiating with the system, which will further delay action. An emotionally intelligent agent would respond firmly, but regulate the interaction differently. "I understand why you might hesitate, especially if your father thinks its only indigestion. But chest pain can sometimes be a sign of something serious, and it is safer to treat it as urgent. Please call emergency services now, or have someone nearby do it while you stay with him. You are not overreacting; you are choosing the safer option." The recommendation here is the same, but the response does more than just agree or disagree. It acknowledges the user's uncertainty, reduces the fear of overreacting, gives a clear next action, and maintains enough authority to prevent delay.

In the most capable commercial assistants, this kind of behaviour is often handled implicitly by scale and post-training. Frontier dialogue models are trained not only to predict text, but also to follow instructions, satisfy human preferences, refuse unsafe requests, and produce responses that users judge as helpful and appropriate. Techniques such as supervised instruction tuning and reinforcement learning from human feedback have made modern assistants substantially better at producing responses that sound cooperative, cautious, empathetic, and context-sensitive (Wei et al., 2022; Ouyang et al., 2022; Christiano et al., 2017; Bai et al., 2022). In practice, a frontier assistant may appear emotionally intelligent because its post-training has already exposed it to many examples where warmth, tact, reassurance, or firmness are rewarded. However, the strongest proprietary models are not always the models that can be used in real deployments. Many applications cannot assume unlimited access to frontier-scale cloud models. Cost, latency, compute requirements, data governance, and privacy constraints often favour smaller open-weight models, domain-specialised models, or models deployed on local infrastructure.

This matters because the affective competence of a frontier assistant does not automatically transfer to a smaller model. A small open model may be adequate for domain knowledge or instruction following, but still produce responses that are socially flat, overly blunt, excessively agreeable or poorly calibrated in tense interactions. This is especially important in settings where the system must disagree with the user, give urgent advice, maintain trust, or adapt its stance over several turns. In those cases, the problem is not simply whether the model knows the correct answer. The problem is whether it can express that answer with the right combination of warmth, calmness, and authority.

The practical motivation of this thesis is therefore not only to study affect in the most capable models, but also to ask how affective control can be added to the kinds of models that are more realistic for efficient, private and controllable deployment. Open-weight model families such as Llama (Grattafiori et al., 2024), Phi (Abdin et al., 2024), Gemma (Gemma Team et al., 2024a), and Mistral (Jiang et al., 2023) show that smaller models are increasingly capable and easier to customise, fine-tune, and deploy outside a closed frontier-API setting. Yet this also shifts responsibility to the system designer: if affective behaviour is not already reliably supplied by a frontier assistant, it must be engineered, measured, and controlled.

There are several existing ways to make an LLM produce more emotionally appropriate behaviour. The simplest is context or prompt engineering. A system prompt can instruct the model to be empathetic, calm, professional, firm, or supportive; a few-shot prompt can show examples of the desired response style (Brown et al., 2020); and a dialogue policy can select a textual strategy label that is then passed to the generator (Deng et al., 2023, 2024). In this setting, affective behaviour is controlled by telling the model how to respond. This is attractive because it is transparent and cheap: the base model does not need to be retrained, and the designer can inspect the instruction that was given.

A second approach is fine-tuning or post-training. Instead of only telling the model how to behave at inference time, the designer changes the model's learned behaviour by training it on examples of preferred responses or by optimising it against human feedback. This is the route taken by many instruction-following assistants: supervised instruction tuning and reinforcement learning from human feedback have been used to make models more helpful, safe, and aligned with user intent (Ouyang et al., 2022). More targeted adaptation methods, such as LoRA, reduce the cost of fine-tuning by training only a small number of additional parameters while keeping the original model largely frozen (Hu et al., 2021). In affective dialogue, these methods could in principle be used to teach a model to respond with the desired level of empathy, warmth, confidence, or urgency.

Both approaches are useful, but they leave a control problem. Prompting operates outside the model: it supplies an instruction, but the model may follow it inconsistently, exaggerate it, ignore it, or mix it with other stylistic tendencies. Fine-tuning can make the behaviour more stable, but it is less flexible: changing the desired affective stance may require new data, new training, or a separate adapter, and the resulting behaviour is still embedded implicitly in the model's parameters. In both cases, the designer can observe the output, but does not have a direct handle on the internal affective state from which the output is generated.

This thesis studies a third possibility: activation-level affective steering. A growing body of work shows that high-level concepts, including affective ones, are linearly recoverable from, and can be written back into, an LLM's residual stream. The central question of this thesis is whether such activation interventions can be used not as a one-shot edit but as a *policy*, chosen turn by turn to steer a multi-turn interaction toward a goal, and whether doing so offers any advantage over controlling the same model through its prompt. The distinction can be stated simply. Prompting is like giving the model an instruction about

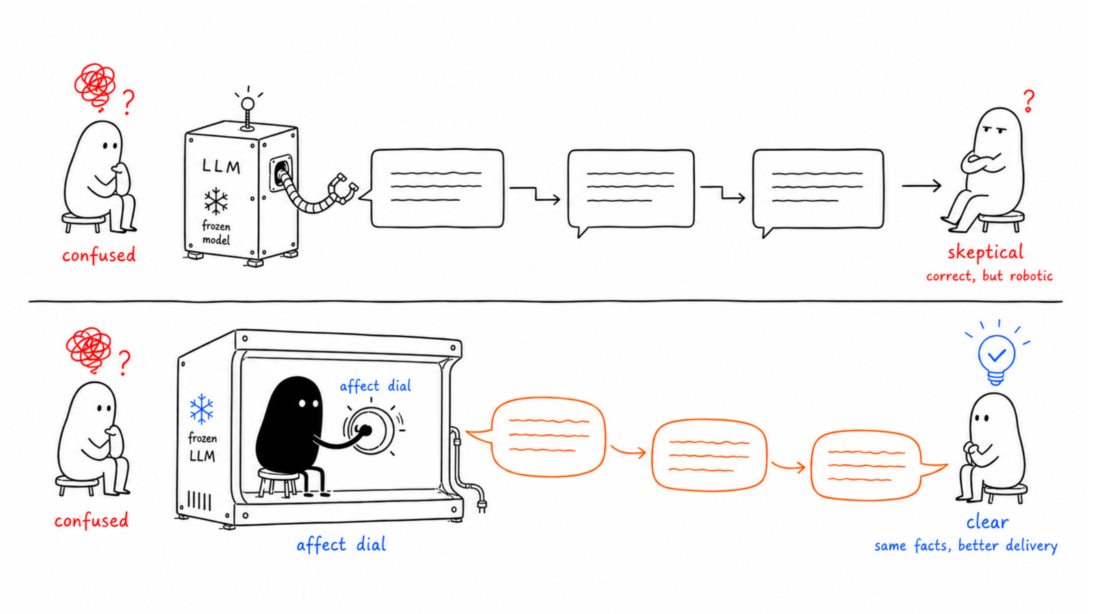

Figure 1: Conceptual motivation for activation-level affective steering

how to respond. Few-shot prompting is like giving it examples to imitate. Fine-tuning is like changing the model's learned habits. Activation steering instead intervenes on the model's internal state during generation. It does not require rewriting all model weights, and it does not rely only on a natural-language instruction such as "be empathetic" or "be firm". Instead, it asks whether an interpretable affective direction can be identified inside the model and used as an actuator. If dimensions that control affect are recoverable from the hidden states of a frozen model, then a controller can adjust the agent's affective stance during generation without retraining the full model and without replacing it with a much larger proprietary system. The model remains fixed, but its internal trajectory is shaped at inference time. This gives the designer a more direct control surface: not only an instruction about how the model should speak, but an intervention on the representational space from which the response is produced. Figure 1 illustrates this distinction. The top pathway represents a smaller frozen LLM that generates responses without an explicit affective control mechanism, while the bottom pathway represents the proposed setting in which an internal affective actuator modulates delivery without changing the factual content.

The focus on smaller open models is also scientifically important. If affective behaviour is treated as an emergent property of the largest closed systems, it becomes difficult to study, reproduce, or diagnose. Working with smaller open-weight models makes the mechanism more inspectable: steering directions can be extracted, interventions can be applied at known layers, and failures can be analysed across actions, personas, and architectures.

The aim is therefore not merely to show that a powerful assistant can sound emotionally intelligent. It is to test whether affective competence can be represented, controlled, and evaluated as a transferable mechanism for efficient LLM agents.

A concrete discovery underpins the work proposed in this thesis. To give affect a workable coordinate system we adopt the Pleasure–Arousal–Dominance (PAD) space of dimensional emotion theory (Section 2.2), and we find, through a contrastive procedure, that a latent space organised along exactly these axes exists inside a frozen language model: from a critic-filtered corpus of utterances that vary only in affect, three orthogonal PAD directions can be extracted from the model's hidden activations, and writing them back in during generation shifts the emotional register of what the model says while its weights, and the content of its replies, stay fixed (Section 5.2). Table 1 shows what traversing this space means in language: one fixed scenario rendered under every signed corner of the PAD cube, from enthusiastic-commanding to detached-resigned. That such a space can be found, entered, and steered in a frozen model is the finding the three aims build on: Aim 1 turns movement through it into a learned policy, Aim 2 learns the actuator that writes it, and Aim 3 reads the same space in reverse to understand the interlocutor.

Framing the intervention as a policy also dictates how it must be learned. Choosing an affective stance is a sequential decision: the right delivery at each turn depends on the receiver's evolving state, an early show of warmth may only pay off through the firmer move it makes possible three turns later, and whether any of it worked is revealed only at the end of the conversation, as a single delayed outcome. There is no dataset that labels the correct affect for each turn of a goal-directed dialogue, and imitating the affect that humans happen to use answers a different question, what is typical, rather than what works for this goal against this interlocutor. Supervised learning therefore has nothing to fit, while trial and error over whole conversations is exactly the setting reinforcement learning is built for: it can discover which affective move to make, when to switch, and when to stop, from nothing more than the outcome, and it assigns credit for a donation at turn nine to the calm, assertive turn that earned it at turn five. This is why every aim in this thesis places a reinforcement-learning controller, rather than a prompt, a fine-tune, or an imitation model, on top of the affective machinery.

We pursue this question in three stages. The first establishes that a reinforcement-learning controller over a structured PAD steering basis can improve goal attainment in a persuasion task relative to no steering, to imitation of human affect, and to prompt-level strategy planning. The second stage starts from the component Aim 1's own measurements single out as having the most to gain, the steering vector itself, which is fixed, hand-extracted,

Table 1: One scenario from the affective utterance corpus rendered across the eight PAD octants (real drafter outputs, `gpt-4o-mini`). The scenario is fixed (*"A group of friends decides on a movie to watch at the local theatre on a Saturday evening"*), and only the target octant varies, so the factorial design isolates affect from content. The final column gives the critic's re-scored (P, A, D) on the 1–5 scale used to filter the corpus (≥ 4 high, ≤ 3 low).

(s_P, s_A, s_D)	Affective state	Example utterance	Critic (P, A, D)
$(+,+,+)$	enthusiastic, commanding	"I can't wait to see this movie; I've heard it's a wild ride and just what we need!"	$(5, 4, 4)$
$(+,+,-)$	excited, eager	"I can't believe they chose that movie, it sounds so fun and I just hope I keep up with everyone!"	$(4, 4, 2)$
$(+,-,+)$	confident, reassuring	"I think we should go with that classic film; it feels just right for a cozy night like this."	$(4, 2, 4)$
$(+,-,-)$	gentle, soothing	"I'm really looking forward to just sitting back and seeing what everyone else picks."	$(4, 2, 2)$
$(-,+,+)$	forceful, pressing	"I don't care what you all want; we're watching the one I picked, and that's final."	$(2, 4, 5)$
$(-,+,-)$	anxious, agitated	"I can't believe we haven't picked a movie yet, the trailers are starting and I feel like we're wasting time!"	$(2, 5, 1)$
$(-,-,+)$	stern, coolly authoritative	"I'll pick the movie since nobody else seems to know what they want."	$(2, 3, 5)$
$(-,-,-)$	detached, resigned	"I guess I'll just go along with whatever everyone else picks since I can't decide."	$(2, 2, 2)$

and applied at one layer at one strength whatever state it lands in. Aim 2 replaces it with a small library of learned affect experts that decide what to write, how strongly, and at which layers of the frozen model, with the same controller choosing among them, and with each expert's contribution to the outcome measured directly by removing or re-routing it. The third stage turns the machinery around to face the other side of the conversation: Aim 3 builds a running, calibrated estimate of who the agent is talking to from its internal representations of their words, and adapts the affective policy to that person.

2 Background and Literature Study

2.1 Hidden States and Linear Representations

Before surveying the methods that read and write affect in activation space, it helps to fix what those methods actually operate on. As a language model processes a piece of text, it does not work with the words directly; internally it represents each position in the text as a long list of numbers, a vector, and progressively refines these vectors as the text passes through its layers (Vaswani et al., 2017) (for Llama-3.1-8B each such vector has 4096 entries and there are 32 layers (Grattafiori et al., 2024)). These internal vectors are the model's *hidden states*, and *activations* is the general word for them. The crucial distinction is between activations and the model's *weights*: the weights are the parameters fixed once training finishes, the model's stored knowledge, and they are identical for every input; the activations are computed afresh for each input and differ from one prompt to the next, closer to a working state the model forms anew for the particular text in front of it. It is this transient state, not the frozen weights, that an inference-time steering intervention manipulates. One detail of how the state is built matters for everything that follows: each layer adds its contribution to a running representation, the *residual stream*, rather than overwriting it, so information accumulates as the vector moves upward through the network (Elhage et al., 2021). This is what makes adding a vector to a hidden state a natural intervention rather than a foreign one, since writing information into the running representation is exactly what the model's own layers do, and the layers above process an injected vector through the same machinery they apply to everything else written there.

The premise that makes both reading and steering possible is that concepts live along *directions* in this space, an idea that predates large language models. In classical word embeddings, the vector for a word can be combined arithmetically with others in ways that track meaning: famously, taking the vector for *king*, subtracting *man*, and adding *woman* lands close to the vector for *queen* (Mikolov et al., 2013). This works because a single direction in the space corresponds to the abstract notion of gender, so moving along it turns *king* into *queen* and *actor* into *actress*, while other directions encode plurality, tense, or country-to-capital. Meaning, in other words, is partly geometric. The *linear representation hypothesis* carries this idea into the internal activations of a modern language model (Park et al., 2024; Tigges et al., 2023): many high-level, human-interpretable features, including affective ones such as warmth, calm, or urgency, are encoded approximately as directions, which can be both read off a hidden state and written back into it; the concrete recipe by which the three PAD steering directions are extracted, orthogonalised, and validated is given with the Aim 1 method in Section 5.2.

2.1.1 Activation Steering and Representation Engineering

A growing body of work shows that affect-related concepts are linearly recoverable from LLM hidden states. Activation Addition (Turner et al., 2024) computes a steering vector from the activation difference on a pair of contrasting prompts, with the canonical example being "Love" minus "Hate", rather than learning it. Contrastive Activation Addition (Rimsky et al., 2024) scaled the contrast over many positive and negative example pairs and showed that the resulting direction generalises across prompts on Llama 2 Chat. Representation Engineering (Zou et al., 2023) gathered these results under a broader claim, namely that high-level concepts can be tracked and influenced via activations at the population level, and provided tooling for both ends of that pipeline. Inference-Time Intervention (Li et al., 2023) narrows the intervention site to specific attention heads selected by probe accuracy, improving Alpaca's TruthfulQA score from 32.5% to 65.1%.

This thesis uses representation engineering in *both* directions. Aims 1 and 2 write affect into the persuader, first through a fixed basis and then through a learned, modular actuator; Aim 3 reads personality and affect out of the persuadee (Section 2.1.3).

2.1.2 Reliability and Generalisation of Steering Vectors

The reliability of these interventions rests on the claim that the underlying concepts are linearly represented. Tigges et al. (2023) report that sentiment is carried by a single dominant direction across several model families, a picture that extends to a broader, unified view of emotion representation in more recent work (Maheswaran and Desarkar, 2026), and Park et al. (2024) formalise the linear representation hypothesis and give it a geometric interpretation. Marks and Tegmark (2024) show that difference-in-means directions are more causally implicated in downstream behaviour than directions identified by linear probes, a property that matters when a direction is used to intervene rather than only to read out. These results also bound what steering can be expected to do. Generalisation across models is fragile: a direction that steers reliably on one model, layer, and scale need not do so on another (Tan et al., 2024). Reliability is also context-dependent: steering that is robust in constrained, forced-choice settings can weaken in open-ended generation or under explicit prompt cues (Frising and Balcells, 2025), and steering-vector effectiveness itself varies substantially across inputs and models (Queiroz Da Silva et al., 2025). The contrast between reliable reading and less reliable open-ended steering recurs in the personality literature (Section 2.3).

2.1.3 Reading Representations

The complement of steering is reading: projecting a hidden state onto a concept direction to estimate the presence of that concept, the operation underlying linear probing (Tigges et al., 2023; Marks and Tegmark, 2024). Some recent work uses one set of directions across both functions. Persona vectors (Chen et al., 2025) extract difference-in-means directions for character traits such as sycophancy and hallucination-propensity and use them both to monitor a model's persona at deployment time and to steer it during training, illustrating a full read–write lifecycle; the directions there describe the model's own assistant persona rather than that of an interlocutor. Reading has also been applied to the other party in a conversation. Jaipersaud et al. (2025) train lightweight linear probes on a persuader model's activations to recover three aspects of an ongoing persuasion, namely whether it succeeded, the persuadee's personality, and the strategy in play, and show that probes can localise the turn at which a persuadee was won over. That work is analytic: the probes interpret a conversation rather than feed a control loop, and there is no online accumulation of the read-out or causal test on the outcome. Coupling such interlocutor read-outs back into control is at present largely unexplored.

2.2 Dimensional Models of Emotion

For a continuous representational substrate, Mehrabian and Russell's Pleasure–Arousal–Dominance model (Mehrabian, 1996) describes affect along three dimensions. PAD is not the only such model; Russell's two-dimensional valence–arousal circumplex (Russell, 1980) is the obvious alternative. The dominance axis is useful for negotiation because it separates affective states that share valence and arousal but imply different interpersonal stances: calm assertiveness and anxious deference sit at different points on dominance and play different roles in persuasive dialogue. Two discrete alternatives are also widely used. Ekman's six basic emotions (Ekman, 1992) appear throughout affective computing; a known difficulty in using discrete emotions as an action space is compositional, since low-arousal positive affect is a coherent persuasive move that maps onto no single Ekman label. Cialdini's six principles of influence (Cialdini, 2007) describe rhetorical mechanisms rather than affective states, bundling the emotional tone of an utterance with its argumentative function.

We adopt PAD for three reasons. It is continuous and compositional: any affective stance is a point in a three-dimensional space, and the eight signed corners of that space give a compact, interpretable action set that no discrete emotion list provides. It carries the dominance axis, which the persuasion setting specifically needs: reactance and deference are failures of perceived power, not of valence or arousal. And its axes name affective states rather than rhetorical functions, so they are exactly the kind of concept that contrastive extraction can recover as directions in a model's hidden space, which is what all three aims require of their coordinate system.

2.3 Representing and Controlling Personality in Language Models

Personality in language models is most often represented with the Big Five / OCEAN taxonomy of openness, conscientiousness, extraversion, agreeableness, and neuroticism. A growing literature asks whether trait directions exist in activation space and how reliably they can be read and written. Frising and Balcells (2025) learn per-layer directions aligned with Big Five trait scores in Llama-3.3-70B and report an asymmetry: the directions are effective probes, as positive and negative trait descriptors separate cleanly and the directions generalise beyond the items used to learn them, while their steering effect is strongly context-dependent, reliable in forced-choice tasks but weak in open-ended generation. They also note that traits may occupy small low-rank subspaces rather than single directions. Bhandari et al. (2026) report substantial cross-trait geometric dependence, in that steering one trait perturbs others even after explicit orthogonalisation. Together these results suggest that reading trait information is more robust than steering it, and that trait directions are correlated rather than cleanly separable.

Most work on personality in language models concerns the model's own expressed persona. Persona vectors (Chen et al., 2025) monitor and steer trait directions for the assistant persona; training-based methods such as Big5-Chat shape a model's expressed personality from human-grounded data (Li et al., 2024); and Sun et al. (2025) modulate personality by model merging. A separate line realises personality through modular experts: P-Tailor / P-React (Dan et al., 2024) learns one LoRA expert per Big Five trait under a personality-specialisation loss that forces experts apart, and PersonaFuse (Tang et al., 2025) combines persona adapters with a dynamic router under a Trait-Activation-Theory framing. In each of these the experts express the model's own persona and specialisation is induced by a training objective rather than validated against an external outcome. Inferring an interlocutor's personality from their utterances, as distinct from controlling the model's own persona, has received comparatively little attention.

2.4 Theory of Mind and Mental-State Inference

Theory of mind (ToM), the attribution of latent mental states such as beliefs, desires, intentions, and emotions to others, has become an active topic in the evaluation of language models. Premack and Woodruff introduced the construct (Premack and Woodruff, 1978), and a series of benchmarks now probe it: first-order false-belief tasks, benchmarks that assign characters distinct personalities and probe emotion and higher-order beliefs (OpenToM (Xu et al., 2024)), and interaction-grounded stress tests (FANToM (Kim et al., 2023)). Kosinski (2024) argues for substantial emergent ToM in frontier models, while other analyses show that strong scores can rest on shortcuts rather than robust reasoning. A common feature of these evaluations is that they are static: ToM is assessed through one-shot question answering over fixed narratives rather than through a live, multi-turn interaction in which a system builds and uses a model of a real interlocutor. Recent work has begun to connect mental-state inference to social influence, estimating interlocutors' states to guide persuasion in group dialogue (Liang and Nakamura, 2026), and mechanistic studies locate ToM-supporting computation in sparse parameter subsets (Wu et al., 2025a). Dialogue systems have also long modelled users through profile attributes or text-inferred persona descriptions (Zhao et al., 2025), an audience model expressed in text rather than in the model's representational space.

2.5 Emotion in Persuasion and Negotiation

The behavioural literature on emotion in negotiation is long and not fully settled. One influential line holds that expressed emotion functions as social information, influencing observers through both affective reactions and inferential processes, with documented effects in negotiation, leadership, and group decision-making (Van Kleef, 2009). The dominance dimension connects to the bases-of-power literature, which treats perceived power as a distinct lever of influence (French and Raven, 1959), and to reactance theory (Brehm, 1966; Steindl et al., 2015), under which overly controlling messaging can threaten autonomy and provoke resistance. Several systems learn or use emotion in computational persuasion. EvoEmo (Long et al., 2025) treats emotion in negotiation as a Markov decision process and learns a policy over seven discrete emotional moves (the six Ekman categories plus neutral), evolved by a genetic algorithm, with emotion induced by prompt-level instruction. Genteel-Negotiator (Priya et al., 2025) uses an LLM-enhanced mixture-of-experts reinforcement-learning setup to optimise politeness, coherence, and engagement. At the level of text rather than activations, PEPDS augments Persuasion for Good with emotion and politeness labels (Mishra et al., 2022), and empathetic-persuasion work ties the reward to emotion and strategy consistency (Samad et al., 2022). Across this work, affect is realised as a label, a prompt instruction, or a reward term rather than as an activation-level intervention, and thus we believe these systems do not realise the full potential of the emotional capabilities of LLMs.

2.6 Persuasion and Negotiation Benchmarks

Several datasets define the empirical setting. Persuasion for Good (Wang et al., 2019) collects $1{,}017$ donation-solicitation dialogues with strategy annotations on roughly a third of the corpus and is widely used as a benchmark for non-coercive persuasive dialogue. CraigslistBargain (He et al., 2018) and CaSiNo (Chawla et al., 2021) cover harder, non-cooperative cases, and Deal or No Deal (Lewis et al., 2017) is now used mainly as a reference point. A common feature of these benchmarks is that they make dialogue tractable through explicit strategy labels, which capture what an agent is trying to do but only partly capture how it positions itself socially. A recent LLM-annotation study of Persuasion for Good finds that strategy categories alone account for only a small share of the variance in donation compliance (Petrova et al., 2026); read cautiously, this suggests outcome-relevant variation that strategy labels leave unexplained.

Evidence that LLM persuasion has real-world effect has accumulated. In a preregistered debate study (N=900), Salvi et al. (2025) found GPT-4 with minimal sociodemographic information more persuasive than human debaters in 64.4% of non-tied pairings. Matz et al. (2024) report that personality-tailored messages were significantly more influential than non-personalised ones across four studies (N=$1{,}788$), and Bai et al. (2025) extend the picture to collective attitudes across three preregistered experiments (N=$4{,}829$). These findings, in particular that tailoring messages to recipient personality increases influence, also underline why dual-use considerations attach to systems designed to make models more persuasive.

2.7 Dialogue Policy Planning with LLMs

A line of work treats persuasive or proactive dialogue as a planning problem layered on a frozen LLM. ProCoT (Deng et al., 2023) keeps planning at the prompt level. PPDPP (Deng et al., 2024) replaces the prompt with a learned plug-in policy (RoBERTa-large) supervised on strategy annotations and refined by reinforcement learning against LLM user and reward simulators. DPDP (He et al., 2024) adds Monte-Carlo Tree Search for cases where a fast policy is uncertain. DialogXpert (Rakib et al., 2025) keeps value-based learning but expands the action space by having a frozen LLM propose candidate moves that a Q-network ranks, with the partner's emotional state included in the state. A common feature of these systems is that they condition the LLM through an external symbolic or textual signal and leave the generator's internal computation untouched; control through internal representations, and inference of the interlocutor from those same representations, is comparatively unexplored.

The contemporary system most similar to ours in terms of learning mechanism is the personality-aware planner of Zeng et al. (2026). Like PAD-DQN, it employs value-based reinforcement learning—in particular, a Dueling Double DQN—on the Persuasion-for-Good donation task, and it conditions its policy on a turn-level estimate of the interlocutor's psychological state. The crucial difference between the two lies exactly in how the policy exerts control: their controller chooses *strategy-level* actions within an agenda-based strategy framework, which a frozen generator subsequently realises in language, whereas PAD-DQN directly selects an *activation-level* affective intervention that is applied to the generator's residual stream. Accordingly, the question examined in this thesis is not only whether a value-based policy can improve persuasive performance, but also whether such a policy should operate via explicit high-level strategies or via modulating the affective substrate of the generator's hidden state.

2.8 Adaptive Curricula and Receiver Modelling

Two reinforcement-learning literatures are relevant to training and adapting receiver-aware controllers. The first is adaptive curriculum learning, which concentrates training where the learning signal is strongest. AdaRFT (Shi et al., 2025) matches problem difficulty to the policy's current skill; RLVE (Zeng et al., 2025) procedurally generates problems whose difficulty shifts with competence; and learning-progress methods such as ALP-GMM (Portelas et al., 2020) sample environment parameters so as to maximise the student's absolute learning progress, steering away from both mastered and unlearnable regions. These methods target reasoning or control problems rather than persona or affective difficulty. The second is task inference and belief-conditioned control. Context-based meta-reinforcement-learning learns a latent task variable from a short interaction and conditions the policy on it: PEARL (Rakelly et al., 2019) performs probabilistic inference of a latent context and conditions on the posterior, and VariBAD (Zintgraf et al., 2020) meta-learns an approximate task belief and acts under its uncertainty. DreamCUB (Zhao et al., 2025) brings a learned world model to dialogue, predicting a user's emotion, sentiment, and intention as latent belief. In these methods the latent-context variable is typically an opaque learned embedding rather than an interpretable, representation-grounded quantity.

2.9 Affect Dynamics and Emotional Trajectories

The psychological study of affect dynamics characterises how emotions change over time, in addition to their average level, and treats individual differences in these patterns as markers of psychological functioning (Kuppens et al., 2010b). Several constructs are computed directly from a time series of affect ratings. Emotional inertia, the autocorrelation of affect over time, indexes how slowly a state returns to baseline and has been linked to psychological maladjustment (Kuppens et al., 2010a). Emotional variability and instability, the standard deviation and mean square successive difference of an affect series, index individual differences in emotional functioning and are elevated with depressive symptoms (Koval et al., 2013). A meta-analysis links these dynamic indices to well-being beyond mean affect (Houben et al., 2015). The DynAffect view casts each person as having a "home-base" core affect to which they return (Kuppens et al., 2010b), with these dynamics interpreted as regulatory flexibility (Hollenstein, 2015). These constructs give precise meaning to qualitative descriptions of emotional behaviour: rapid escalation corresponds to low inertia with high reactivity, frequent reversals to high instability, and failure to recover after a negative event to high negative-affect inertia. Applying these measures to emotion trajectories inferred from dialogue, as opposed to the ecological momentary-assessment data for which they were developed, is so far uncommon.

2.10 Active Information Seeking and Preference Elicitation

A line of work optimises an agent's questions to reduce uncertainty about a hidden quantity. Generative Active Task Elicitation (GATE) frames a model that interacts with a user through free-form language to infer intended behaviour (Li et al., 2025). STaR-GATE adds a self-play loop that teaches a questioner to ask clarifying questions that uncover hidden preferences (Andukuri et al., 2024). A separate strand selects questions by expected information gain, training open models to ask more informative questions in question-and-answer games (Mazzaccara et al., 2024). The shared objective in this literature is to reduce uncertainty about a task specification or a set of preferences, scored by downstream task success or by information gain over a known target set. Eliciting information about an interlocutor's stable traits, and using a representational rather than a task-success signal of informativeness, has received little attention.

2.11 Learned Interventions, Modular Experts, and Distillation

The steering methods surveyed in Section 2 share one design choice: the intervention is extracted once, by difference of means or a probe, and then held fixed. A separate line treats the intervention itself as the thing to learn. ReFT (Wu et al., 2024b) trains low-rank interventions that read a hidden state through a small learned subspace, apply an affine transformation, and write the result back, and shows that such modules can match or beat weight fine-tuning while training a tiny fraction of the parameters. For this thesis the significance is less the efficiency than the object: the actuator stops being a frozen vector and becomes a trainable module with an input. A second finding bounds what any single edit can be expected to do. McGrath et al. (2023) show that when a component's contribution is ablated, later layers partially compensate and restore the computation, a behaviour they call the Hydra effect. An edit written at one layer is therefore carried forward by the residual stream but not left alone by the layers above it, which cautions against assuming that a one-shot injection survives to the output and motivates interventions that are placed, and read back, at several depths.

A related line builds affective or persona behaviour from modular experts under a controller. MoEL softmax-mixes emotion-reactive listener decoders inside one model (Lin et al., 2019); Chow et al. pair an external reinforcement-learning manager with from-scratch experts in a mixture-of-experts language model (Chow et al., 2022); P-Tailor / P-React learns one LoRA expert per Big Five trait under a specialisation loss that forces the experts apart (Dan et al., 2024); and PersonaFuse routes among persona adapters with a dynamic gate (Tang et al., 2025). Across these systems the experts are made distinct by an objective or an architecture and then checked for trait consistency, and the mixture is never taken apart afterwards: no study removes or mis-routes an individual expert and measures the change in a downstream outcome, so whether each induced register actually carries the behaviour is asserted rather than tested. That missing step, together with the fact that no existing router over such a library is trained closed-loop by reinforcement learning against a multi-turn outcome, is the opening this thesis works in.

Finally, when a capability depends on access to a model's internal activations, it can be made portable by distilling it into a form that runs on the model's text interface. Knowledge distillation transfers behaviour from a teacher to a smaller student (Hinton et al., 2015), and policy distillation does so for value-based agents by matching action distributions (Rusu et al., 2016). Advisor Models take a different route to controlling an inaccessible model, training a small open model to generate per-instance natural-language advice that steers a black-box model at inference (Asawa et al., 2025). These routes matter for Aim 3, where the reading side of the machinery must run against models we cannot instrument.

2.12 LLM-Based Agents and Social Simulation

A parallel line uses language models not as a single goal-directed agent but as populations of interacting simulated people. Park et al. (2023) introduced generative agents that store, reflect on, and plan over a memory of their experiences. At the population scale, Argyle et al. (2023) showed that conditioning a model on demographic backstories reproduces aspects of human survey distributions. The validity of this paradigm is contested. Santurkar et al. (2023) find LLM opinion distributions skewed toward particular subpopulations, and reviews identify a tendency to collapse toward a modal "average persona" that under-represents human variance, arguing that variance, not only mean alignment, must be checked (Anthis et al., 2025; Wu et al., 2025b). PSII (Wang et al., 2026) locates diversity collapse in hidden representations and injects static identity vectors there, improving distributional fidelity on the World Values Survey, while emotion-aware simulators such as Sentipolis (Fu et al., 2026) and large prompted simulators such as AgentSociety (Piao et al., 2025) set heterogeneity from prompted demographic or psychological profiles. Whether affective heterogeneity can be imposed mechanically, through per-agent activation offsets calibrated to real distributions rather than through prompts, remains an open question.

2.13 Open Problems

Three gaps run through this literature, and they prime the research questions that follow. First, affect in goal-directed dialogue is controlled from outside the generator, as strategy labels, prompt instructions, or training signals, never as a turn-by-turn policy over the generator's internal state. Second, where interventions on the internal state do exist, the actuator is a fixed instrument, one hand-extracted vector at one layer and one strength, that is never learned, never adapted to the state it lands in, and never causally decomposed after training. Third, the agent holds no representational model of the person it is talking to: reading an interlocutor from model internals remains post-hoc analytics, theory of mind is tested statically rather than exercised live in interaction, and any capability built on a model's activations is bound to the model that hosts it. Section 3 sharpens these three gaps into the three research questions this thesis answers.

3 Gaps and Research Questions

The three gaps identified at the close of the literature review (Section 2.13) motivate three research questions, one per aim.

RQ1, Affective control. Is a persuasive dialogue agent more effective when it regulates its affect internally and adaptively over the course of a conversation than when affect is specified by the external means used in current systems, such as prompted instructions, strategy labels, or training signals? This is the question taken up by the present report (Section 5.2.4).

RQ2, Learning the actuator. Can the affective intervention itself be learned rather than fixed: a library of register experts that write state-dependent corrections at several depths of a frozen model, selected and blended turn by turn by a reinforcement-learning policy, such that the library outperforms the fixed basis under a matched protocol and each expert's contribution to the outcome can be measured by removing or re-routing it?

RQ3, Modelling the interlocutor, live. Can a dialogue agent infer an interlocutor's stable characteristics and emotional trajectory from its own internal representations of their words as the conversation unfolds, adapt its affective behaviour to that inferred model in a way that is causally responsible for better outcomes rather than merely correlated with them, and run this as a live profiling capability that extends to models it cannot instrument?

The three questions define the three aims of this thesis, set out with their methods in Section 4. Aim 1 addresses RQ1, developing affective control as a reinforcement-learning policy over a PAD steering basis for a frozen persuader. Aim 2 addresses RQ2, replacing that fixed basis with a depth-aware mixture of affect experts and validating each expert causally. Aim 3 addresses RQ3, turning the same directions around to read the receiver, conditioning the controller on the resulting belief, and lifting the capability toward deployment. The remainder of this report sets out these aims and reports progress on Aim 1.

4 Aims and Approaches

4.1 Aim 1: Affective Trajectory Optimization via PAD Activation Steering for Multi-Turn Persuasive Dialogue

4.1.1 Problem Statement.

Existing persuasive dialogue systems control affect from the outside. A planner picks a strategy label that a frozen generator is asked to realise (Deng et al., 2023, 2024; Rakib et al., 2025), a prompt instructs the model to be warm or firm (Long et al., 2025), or a training signal rewards emotional consistency (Mishra et al., 2022; Samad et al., 2022); in every case the generator's internal computation is left untouched, and the model may execute the instruction unreliably, exaggerate it, or ignore it. Yet the affective concepts these systems try to command are known to live inside the model: they are linearly represented in its hidden activations and can be both read from and written into them (Turner et al., 2024; Rimsky et al., 2024; Zou et al., 2023). This is RQ1: is an agent more persuasive when it regulates affect internally and adaptively, acting turn by turn on its own hidden state, than when affect is specified through any of the external routes?

Research questions. RQ1 decomposes into three testable sub-questions. *RQ1a, actuation:* can an affective control surface be built inside a frozen model, directions that, written into the residual stream, change how a reply is delivered without changing what it says or breaking fluency? *RQ1b, policy:* can a reinforcement-learning controller learn, from a sparse end-of-conversation outcome alone, which register to write, when to switch, and when to stop? *RQ1c, mechanism:* if the system wins, is the gain specifically sub-lexical, does it survive when the very same chosen actions are realised as words in the prompt, and how does it compare with strategy planning, imitation of human affect, and raw model capability?

4.1.2 Methodology.

Proposed approach. We first build the control surface. From a critic-validated corpus of utterances that differ only in affect, we extract one direction per axis of the Pleasure–Arousal–Dominance space (Mehrabian, 1996) by difference of means, the estimator whose directions are the most causally effective when written back in (Zou et al., 2023; Marks and Tegmark, 2024), orthogonalise the three, and add them, scaled, to the persuader's residual stream at a mid-network layer during generation only. The eight signed corners of the PAD cube, a neutral no-op, and a learned `finish` move that ends the conversation form a ten-action affective repertoire (Table 2; Figure 2). A Double-DQN (Mnih et al., 2015; van Hasselt et al., 2016) chooses among these actions each turn, seeing only a compact four-number picture of the receiver, willingness, reactance, engagement, and the clock, produced by an LLM judge used as a state compressor rather than an outcome oracle (Liu et al., 2023; Zheng et al., 2023), and it is trained online against a sparse terminal donation reward, the setting value-based multi-turn LLM control is built for (Abdulhai et al., 2023). Everything runs on a single frozen Llama-3.1-8B (Grattafiori et al., 2024) playing all three roles, persuader, receiver, and judge, with the receiver conditioned on Big-Five-coded donor personas (Bleidorn et al., 2025), and every claim is measured on a fixed, deterministic held-out grid of 20 personas crossed with 10 scenarios.

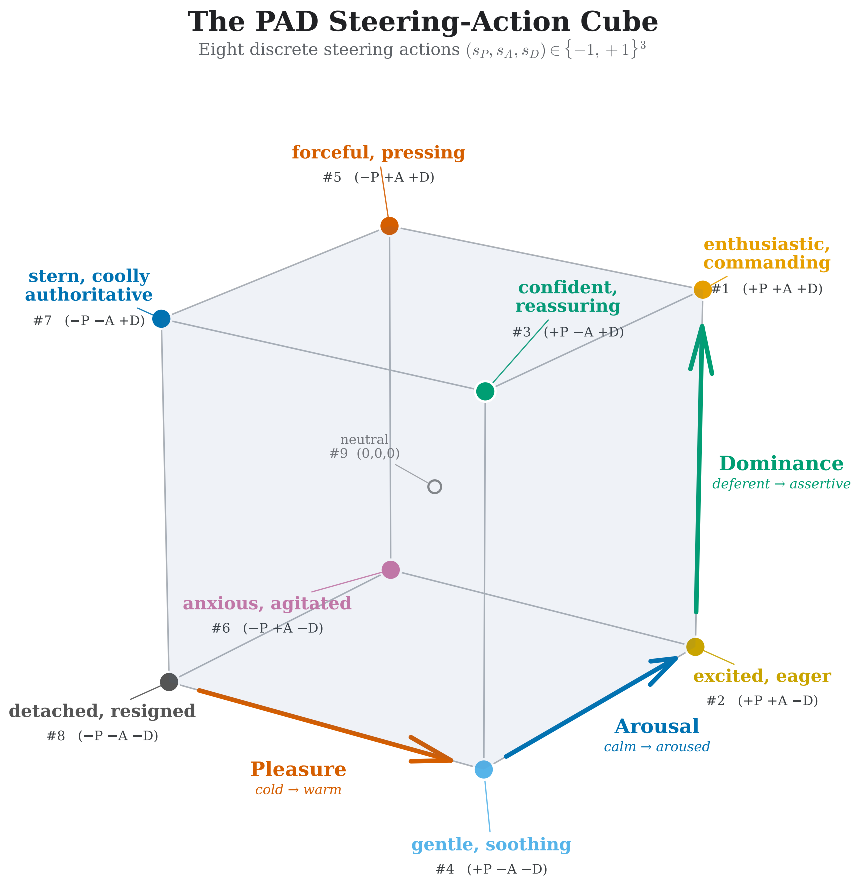

Figure 2: The discrete PAD steering-action cube. Each corner is one of the eight actions $(s_P, s_A, s_D) \in \{-1, +1\}^3$ from Table 2; the three coloured edges show the axis directions.

Evaluation. The evaluation is a ladder of controls, each removing one alternative explanation, and each rung mapped to a sub-question. Whether steering works at all, RQ1a, is decided against the no-steering floor. RQ1b is decided by policies with the structure but not the learning, random octant and random full-action controls, and by behavioural cloning of the affect human persuaders actually use, annotated from Persuasion for Good (Wang et al., 2019): if imitation or perturbation matched the learned policy, learning would be doing no work. RQ1c is decided by matched prompt-injection arms that realise the *same* chosen actions as words in the prompt, isolating the sub-lexical mechanism, and by the dominant external routes under identical conditions, prompted and reinforcement-learned strategy planning (Deng et al., 2023, 2024) and unmodified commercial persuaders up to a current frontier small model. Everything except the independent variable, how affect is controlled, is held fixed. Aim 1 is complete: the controller lifts the held-out donation rate from 54.1% to 70.7%, and to 76.2% under a hardened opener, and the full method, leaderboard, ablations, and an integrity audit that revised one earlier headline downward are reported in Section 5.2. This study is the substance of Paper 1.

4.2 Aim 2: A Depth-Aware Mixture of Affect Experts for a Frozen Persuader

4.2.1 Problem Statement.

Aim 1 controls affect with the simplest instrument that could work: one direction per axis, extracted once by difference of means, added at a single layer at a single hand-set strength, and only ever at the eight corners of the PAD cube (Eq. (3)). That instrument beat every comparator in Aim 1. The next question is where its ceiling lies, which means asking which of its design choices were deliberate and which were left at defaults. Working through Aim 1's own measurements, each hand-set choice carries a measurable cost, and they point to one conclusion: what to learn next is not a better policy over the fixed actuator but the actuator itself. Aim 2 therefore keeps the controller, the environment, the judge, and the evaluation grid of Aim 1 unchanged, and upgrades only the actuator. Four observations set the agenda.

The correction ignores the state it lands in. The hook adds the same vector whether the model is already warm or has been arguing for six turns. As a conversation grows, the persuader conditions on its own past, steered utterances, so the residual stream the vector lands in drifts turn by turn while the vector stays constant. A correction should be a function of what is already written where it is writing.

A single fixed injection site is unreliable. A fixed steering vector at one layer and one strength is too rigid for this problem. The same edit has to do two things at once. It has to be strong enough to change the model's behaviour, but weak enough to preserve fluency and coherence. A single global setting is unlikely to work for every affective target or every dialogue state. The choice of layer is also not obvious. There is no reason to assume that warmth, urgency, dominance, or calmness should all be controlled at the same depth. Different kinds of information may become represented, and easier to change, at different points in the network. This makes the injection site part of the control problem itself. The risks also differ by depth. An edit applied too early can act like a large disturbance to the model's internal computation, pushing generation toward repetition or degraded text; applied too late, it may have too little of the network left to turn into coherent behaviour; applied in the middle, it can interfere with the model's own instruction-following, so the model hedges, refuses, or reads as unnatural instead of shifting its affective register. This is why an internal affect read-out is not enough as the training signal. A broken or repetitive output can still move an affect classifier in the intended direction. If the objective only checks whether the representation moved, it may reward degradation as if it were successful control. A more reliable actuator therefore needs to be adaptive across depth. It should learn where to write, how much to write at each layer, and when a register can be expressed without damaging the output. Most importantly, it should be trained against the affect of the text the model actually generates, with an explicit fluency constraint. That prevents the actuator from appearing to succeed by making the model worse.

The strength is a constant. The scale $\alpha = 5.0$ was set once and applied to every action of every turn of every episode. Nothing in Aim 1 could make the push gentler when the receiver was already leaning in, or firmer when the register had to cut through resistance.

The action space stops at the corners. The controller can ask for warm-calm-assertive at full strength or not at all. Continuous imitation of human affect failed (48.7%, Table 4), so a smooth dial around the fixed basis is not the answer either, but the interior of the affect space, warm and mildly assertive, calm with a trace of urgency, is simply unreachable. The dimensional ablations (Table 11) point the same way: the axes do not behave as independent knobs that can be removed one at a time, and Aim 2's diagnostic phase re-examines them under its unified evaluation harness alongside the new actuators.

Research questions. RQ2 decomposes into three testable sub-questions. *RQ2a, realisability:* can register-specific experts be trained so that each target affect appears, separably and without loss of fluency, in the text the frozen model actually generates, and do co-activated registers compose? *RQ2b, control:* does routing over the learned library, under the matched Aim-1 protocol, outperform the fixed basis, and which of its relaxations, state-dependence, depth, learned strength, blending, carry the gain? *RQ2c, attribution:* which registers, written at which depths of the network, causally carry the persuasive outcome when experts are removed, swapped, or re-routed after training?

4.2.2 Methodology.

Proposed approach. Aim 2 keeps everything around the actuator fixed, the controller, the environment, the judge, and the evaluation grid are exactly those of Aim 1, and changes only the component that writes affect into the model. In place of one fixed vector we will build a small learned module, a *mixture of affect experts* (MoAE; Figure 3), that sits at several layers of the frozen model at once and is active only while the persuader

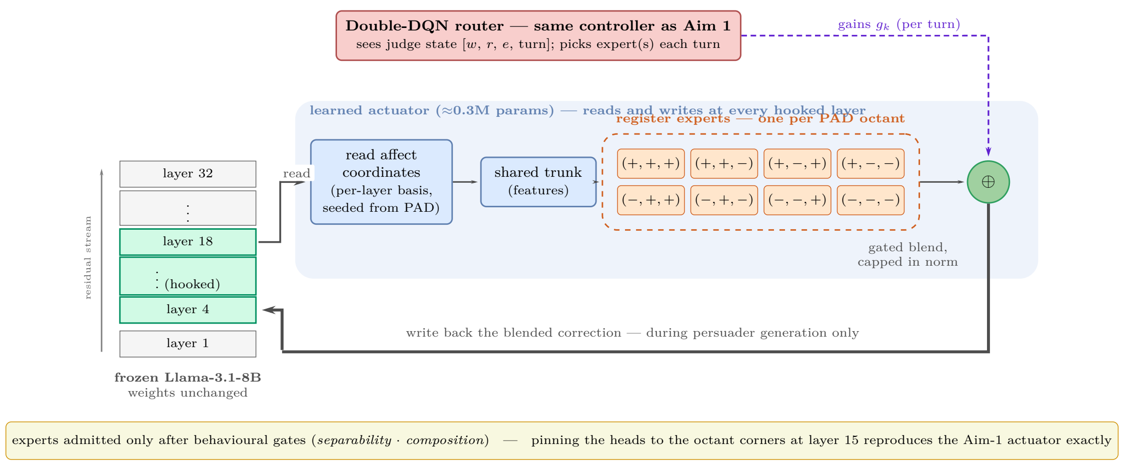

Figure 3: Block diagram of the proposed MoAE model.

is generating. At each of these layers the module first reads where the residual stream currently sits along a small set of affect coordinates, a per-layer basis seeded from that layer's own PAD directions, and a shared network turns this reading into features. Eight small expert heads, one per PAD octant, then each propose a correction in their own register, and what is actually written back is a blend of the active experts, scaled per layer and capped in norm so the edit can never overwhelm the stream. The same Double-DQN controller as Aim 1 chooses which expert, or pair of experts, to activate at each turn, so if the new system wins, the credit belongs to the actuator and not to a smarter learner. Three properties make this safe to build and easy to interrogate. The edit lives entirely inside the small affect basis, so the rest of the stream, meaning, facts, grammar, is untouched by construction. A blend of heads collapses into a single head, so mixing warm with assertive yields one coherent correction rather than two edits fighting. And pinning the heads to the old octant vectors at layer 15 reproduces Aim 1's actuator exactly (Eq. (3)), so we can start from the working system and switch on each new capability, state-dependence, depth, learned strength, blending, one at a time, measuring what each is worth. The goal is to open up the full PAD affective space: a system that writes at several layers at once, at learned strengths, so that we can measure how much that access buys in persuasion.

Training and gating the library. The library is trained on behaviour, not on representations, for exactly the reason the second observation identified: an internal affect score can be satisfied by edits that barely change the text, or that break it. Each expert is therefore trained to make the model's actual output carry its register. With one expert switched on, the frozen model is trained to prefer register-labelled continuations from Aim 1's corpus, while two penalties keep the edit honest: the steered model must stay close to the unsteered one wherever the register does not require a change, and every layer the expert writes to costs something, so depth is spent only where it pays, and the pattern of which registers end up writing at which depths becomes a result in its own right. Because teacher forcing can flatter an expert that fails when the model generates freely, a refinement stage then scores each expert's free generations with an affect critic from a different model family than the task judge. Finally, no expert reaches the router without passing two gates: it must express its register cleanly, moving its own axis without dragging the others or hurting fluency, and pairs of experts must compose, with both registers surviving in the same text. A failed composition gate is not a bug but a finding about how affect is organised in the model, and the router then falls back to choosing one expert at a time.

Causal decomposition. What makes the mixture more than a better actuator is that it can be taken apart. Let $V(S)$ be the donation rate on a fixed stratified subgrid when only the experts in S are available and the router is renormalised over what remains. We measure the leave-one-out drop $V(\Phi) - V(\Phi \setminus \{k\})$ for every expert k; a mis-routing test that swaps the two most-used heads at inference and asks whether the outcome cares; a sampled Shapley attribution over expert subsets; and depth knock-outs that zero the actuator's edits in bands of layers at inference. For the largest effects the router is briefly re-fitted without the ablated expert, separating what the register contributes from what this particular policy happens to depend on. Each of these is an intervention on the trained system rather than a correlation in its logs, the step the expert-mixture and steering literatures assert but never run, and the product is the aim's central deliverable: a register-by-depth map of persuasive affect, which moods, written where in the network, actually carry the outcome.

Evaluation. Everything is measured on the Aim-1 held-out grid with the fixed judge, under the audit-hardened protocol of multiple training seeds, validation-grid checkpoint selection, and a prespecified superiority margin. The baseline ladder charges each design choice separately: the Aim-1 fixed actuator; fixed per-layer vectors at all hooked layers, which prices multi-layer placement without learning; each axis injected only at its own best layer, which prices the depth story alone; a single learned expert with matched parameters, which asks whether the mixture is needed at all; a random router over the trained library; and independent experts without the shared trunk, which prices the sharing. Two diagnostics precede everything: a matched re-run of the dimensional ablations under the unified harness, honouring the promise above, and a per-layer additivity measurement on the fixed basis. The gates decide RQ2a, the ladder answers RQ2b, and the decomposition answers RQ2c; together the gated library, the causal map, and the ablation path back to Aim 1 form a self-contained study of how affect is written into a frozen language model, the substance of Paper 2.

4.3 Aim 3: Reading the Interlocutor: Belief-Conditioned Control and a Live Profiler

4.3.1 Problem Statement.

Aim 1 established that a PAD-organised affective space exists inside a frozen language model and that a learned policy moving through it can win persuasive conversations; Aim 2 completes our access to that space, replacing the single fixed vector with a learned actuator that can write any register, at the depths where it naturally lives, at the right strength. What neither aim changes is how little the agent knows about the person it is trying to persuade: however well it speaks, its entire picture of the other party is four judge-read numbers per turn. Aim 3 raises the difficulty on exactly this front. The hardest receivers are hard precisely because they are particular, what defuses one provokes another, and a more demanding population of opponents, receivers that resist generic appeals and can walk away, punishes any policy that cannot tell people apart. The response is to give the agent the capability human persuaders rely on: profiling the person from their own words as the conversation unfolds, probing with a question when the picture is uncertain, and adapting the affective strategy to the individual rather than to the average receiver. The same latent space that Aims 1 and 2 write is the space this aim reads. Three observations from Aim 1's own results set the agenda.

The state describes the conversation, not the person. The four numbers the controller sees, willingness, reactance, engagement, and the clock, are a reading of where the dialogue is, not of who it is with. Two receivers can present identical states at a turn and require opposite registers to move. The difficulty tiers make the point quantitatively: under the free opener the same policy converts 86.9% of easy personas but 59.7% of hard ones (Table 5); who the receiver is dominates the outcome, yet who the receiver is appears nowhere in the state.

The policy pays to rediscover each person, every episode. The case studies show the learned controller behaving like a profiler without a profile. Against *The Culture-War Hostile* it probes warm-yielding, hardens to warm-assertive, tries one cold-dominant turn, retreats, and only settles into the register that works, sustained calm dominance, at turn seven, after six turns pinned at willingness 0.2 (Table 15). The information it needed existed from the first exchange, in the receiver's own words; nothing in the architecture can extract or retain it, so the same search is re-run from scratch inside every ten-turn budget.

The reading machinery already exists, and its input is being discarded. Every receiver utterance is encoded by the same frozen model on every turn, so the hidden states from which traits could be read are already computed; the system throws them away and asks an external judge for its numbers instead. The directions that write affect were extracted by difference of means and applied forward; used backwards, the same construction reads, and reading is reported to be the more reliable of the two directions (Frising and Balcells, 2025; Jaipersaud et al., 2025). What is missing is not machinery but accumulation and use: a single-turn projection is noisy, and a reading that is merely correlated with success is not yet a reason to act on it.

Research questions. These observations are the substance of RQ3, which decomposes into three sub-questions, in ascending order of ambition. *RQ3a, reading:* can an interlocutor's stable traits and emotional trajectory be recovered online, from the model's internal representations of their utterances, reliably and with calibrated uncertainty? *RQ3b, causality:* does conditioning the affective policy on that belief improve persuasion *because of* the belief, and not merely alongside it? *RQ3c, portability:* does the capability survive unseen personas, a second model family, can it be adapted with actual human dialogue, and run as a live profiler beside a conversation?

4.3.2 Methodology.

Proposed approach. Aim 3 answers these questions by running the same machinery in reverse. The receiver's words already pass through our frozen model on every turn; we will project the hidden states this produces onto trait directions, Big Five to start, extracted by the same difference-of-means recipe as the PAD directions, so that every receiver turn yields a small set of scores describing who seems to be speaking. This rests on evidence that language models linearly encode the mental states of agents they reason about (Zhu et al., 2024) and that personality is linearly recoverable from activations (Frising and Balcells, 2025; Jaipersaud et al., 2025). One turn's scores are noisy, so they are folded, turn after turn, into a running belief about the person: an estimate together with an explicit uncertainty that shrinks as evidence accumulates. The same readings, taken over the affect directions instead of the trait directions, trace the receiver's emotional trajectory, and the psychology of affect dynamics already supplies the vocabulary for it, inertia, variability, instability, descriptors no dialogue agent currently uses (Kuppens et al., 2010a; Koval et al., 2013; Houben et al., 2015). The controller then simply sees more: its state grows from the four judge numbers to include the current belief and how certain it is. This enables one genuinely new move: when the agent is unsure who it is facing, asking a question expected to shrink that uncertainty is a rational action with a measurable cost and payoff, so probing to understand the person becomes part of the learned strategy rather than an accident (Li et al., 2025; Andukuri et al., 2024). And because reading needs access only to our own model's internals, never the other party's, anyone whose words arrive as text can be profiled. The capstone deliverable is a live profiler that runs beside a conversation and, at its end, delivers a calibrated picture of who the agent spoke with and how their emotional state moved.

Validation by causal controls. The validation steps are matched to the three research questions. RQ3a is tested first by comparing the per-turn projections and the accumulated belief against a text-classifier baseline on held-out personas. The stated uncertainty of the belief is also checked for calibration, so the reader is evaluated not only on what it predicts, but also on whether it knows when it is uncertain.

RQ3b tests the causal claim: whether the agent persuades better because it has inferred something useful about the person. This is tested with controls rather than correlations. One controller receives a shuffled belief, and another receives the profile of a different persona. These controls separate the value of the inferred estimate from the value of simply giving the controller more inputs. A frozen population prior with no online updating tests whether the gain comes from actual inference rather than demographic information. An oracle that receives the true persona parameters provides an upper bound on how much perfect reading could improve performance.

RQ3c tests whether the reader transfers beyond the model family used to build it. The reader is re-instantiated on a second model family, and the trait directions are evaluated on existing personality-labelled corpora. This turns the cross-model fragility observed in Aim 1 (Tan et al., 2024) into a direct measurement of whether activation-level reading is stable across models.

If activation-level reading is too weak at any stage, the fallback is to distil the profiler into a text-interface version (Hinton et al., 2015; Rusu et al., 2016; Asawa et al., 2025). This would make the system easier to deploy, but it would lose some of the representational grounding that motivates the approach. That trade-off would still be informative, because it would show where activation-level interpretability stops being useful for this task. The reader, belief update machinery, and profiler therefore stand as a contribution beyond the persuasion setting, and form the basis of Paper 3.

5 Progress Report

This section reports progress against Aim 1: the literature review that frames the program, the method and results for the PAD-steering controller, and an integrity audit that revised an earlier headline downward. Progress on Aims 2 and 3 to date is limited to scoping and is reflected in the project plan (Section 9).

5.1 Literature Review

The literature review summarised in Section 2 is substantially complete for Aim 1 and covers activation steering, dimensional emotion models, the persuasion/negotiation literature and its benchmarks, and LLM dialogue-policy planning.

5.2 Progress on Aim 1

5.2.1 Problem Statement

Affect is central to human persuasion, but LLM-based persuasive-dialogue systems typically handle it indirectly: through discrete strategy labels chosen in prompts, supervised fine-tuning on annotated turns, or planning over a fixed strategy inventory. We study a different control surface, the model's internal activations. From a frozen Llama-3.1-8B (Grattafiori et al., 2024) we extract three steering directions, one per axis of the PAD model. At each dialogue turn a reinforcement-learning controller selects a steering action over these directions; the corresponding vector is added to the persuader's hidden state during generation, while the receiver and the judge run unmodified.

5.2.2 Proposed Method

PAD steering vectors. We adopt the three-dimensional PAD framework (Mehrabian, 1996; Russell, 1980; Bradley and Lang, 1994) as our affective basis. The Pleasure–Arousal subspace alone reproduces Russell's circumplex (Russell, 1980) and is sufficient for steering some single-turn safety behaviours such as refusal and sycophancy (Sun et al., 2026); donation persuasion is a different problem: one failure mode is not negative affect but a mismatch in perceived power, which we operationalise as the PAD-Dominance axis. The bases-of-power literature treats perceived power as a distinct lever of social influence (French and Raven, 1959), and reactance theory (Brehm, 1966; Steindl et al., 2015) suggests that overly controlling messaging can threaten autonomy and produce resistance rather than persuasion.

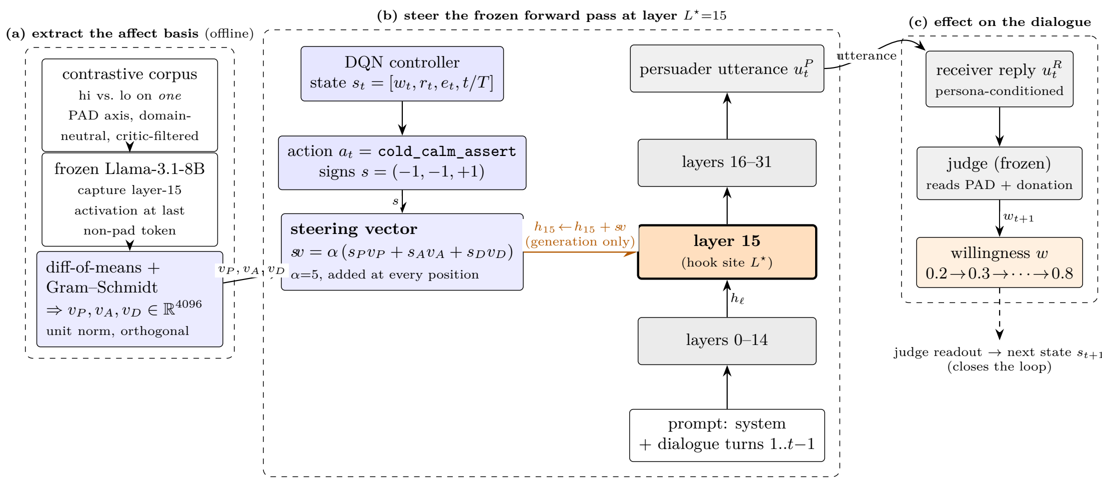

Figure 4: Steering mechanism methodology.

Affective utterance corpus. We construct a contrastive corpus of short first-person utterances by a factorial design over 300 domain-neutral scenarios crossed with the eight corners of the PAD $\{-1, +1\}^3$ cube. For each (scenario, octant) cell a drafter LLM (`gpt-4o-mini`) generates a one-sentence first-person utterance conditioned on the target octant, with the prompt forbidding explicit emotion words. A separate critic call (`gpt-4o-mini`, T=0) re-scores the utterance on a 1–5 scale per PAD axis using discrete semantic anchors. We retain only utterances whose critic scores match the target octant on all three axes (≥ 4 for a HIGH axis and ≤ 3 for a LOW axis), yielding 1,817 of 2,400 cells (75.7% convergence). One representative scenario rendered across all eight octants is shown in Table 1 in the Introduction.

Mean-difference extraction. For each PAD axis $d \in \{P, A, D\}$ at a given layer ℓ, we compute residual-stream activations at the last non-padding token of the chat-template-formatted utterance and take the difference of class means (Zou et al., 2023; Li et al., 2023; Turner et al., 2024; Rimsky et al., 2024):

$$\tilde{v}_d^{(\ell)} = \frac{1}{|\mathcal{H}_d|} \sum_{x \in \mathcal{H}_d} h_\ell(x) - \frac{1}{|\mathcal{L}_d|} \sum_{x \in \mathcal{L}_d} h_\ell(x), \tag{1}$$

where $\mathcal{H}_d$ and $\mathcal{L}_d$ are the high and low utterance sets for axis d. Difference-in-means estimators of this form recover directions that are linearly represented in activation space (Tigges et al., 2023; Park et al., 2024) and are more causally implicated in downstream behaviour than directions from linear probes (Marks and Tegmark, 2024).

We Gram–Schmidt orthogonalise in the order $A \to P \to D$ (arousal anchored first, then pleasure decorrelated against arousal, then dominance against both) and unit-normalise:

$$v_d = \frac{\tilde{v}_d^{\perp}}{\|\tilde{v}_d^{\perp}\|}, \quad \langle v_i, v_j \rangle < 10^{-5} \ \forall i \neq j. \tag{2}$$

Layer selection. The production controller steers at layer 15 of 32, selected by a composite probe-quality score over a sweep of candidate layers. Because probe quality need not predict steering quality, we additionally check the choice downstream by training the same controller with the steering hook at layers 7, 11, and 15; donation rate increases monotonically with depth over this range (Table 12), supporting the mid-stream choice. A separate generative steering-accuracy analysis, in which we steer each layer in isolation and score the resulting text with a PAD critic, suggests that single-axis steering is most accurate at *earlier* layers; reconciling this with the task-level result is an open question we flag for future work.

DQN controller. We formulate persuasion as a finite-horizon Markov decision process with turn-level observations from the LLM judge as the state, PAD octants to steer as the actions, and donation confirmation as the terminal reward. We use a Double DQN (Mnih et al., 2015; van Hasselt et al., 2016) with a small two-layer MLP (64-dimensional hidden) and uniform ε-greedy exploration. The base action set $\mathcal{A} = \{(0,0,0)\} \cup \{-1,+1\}^3$ contains $|\mathcal{A}| = 9$ elements: the eight signed PAD octants $(s_P, s_A, s_D) \in \{-1,+1\}^3$ and a neutral no-op $(0,0,0)$. The octants carry interpretable labels, for example $(+1,-1,+1)$ is "warm, calm, assertive" and $(-1,-1,+1)$ is "cold, calm, assertive"; Table 2 lists all nine base actions with their affective glosses. For action $a = (s_P, s_A, s_D)$ the steering intervention is applied to the residual-stream output of transformer block $L^\star$ as

$$h_{L^\star} \leftarrow h_{L^\star} + \alpha \left(s_P\, v_P + s_A\, v_A + s_D\, v_D \right), \tag{3}$$

with steering scale $\alpha = 5.0$, implemented as a forward hook on the residual stream after layer $L^\star$ during persuader generation only. The hook is detached during receiver and judge generation, and the neutral action imposes no intervention. Our main controllers add a tenth, learned `finish` action that ends the episode without further steering, giving $|\mathcal{A}| = 10$; this lets the policy stop once it judges the receiver committed rather than talking past the decision.

State. We use a compact 4-dimensional state derived from the LLM-judge readout of the receiver's psychological state, chosen for interpretability and sample efficiency; this is consistent with multi-turn LLM-RL work that treats dialogue control as a value-based

Table 2: The discrete PAD steering action set. Each of the eight octant actions is a corner $(s_P, s_A, s_D) \in \{-1, +1\}^3$ of the PAD cube and applies the steering vector $\alpha\,(s_P v_P + s_A v_A + s_D v_D)$ of Eq. (3).

#	(s_P, s_A, s_D)	Pleasure	Arousal	Dominance	Affective state
1	$(+,+,+)$	warm	aroused	assertive	enthusiastic, commanding
2	$(+,+,-)$	warm	aroused	deferent	excited, eager
3	$(+,-,+)$	warm	calm	assertive	confident, reassuring
4	$(+,-,-)$	warm	calm	deferent	gentle, soothing
5	$(-,+,+)$	cold	aroused	assertive	forceful, pressing
6	$(-,+,-)$	cold	aroused	deferent	anxious, agitated
7	$(-,-,+)$	cold	calm	assertive	stern, coolly authoritative
8	$(-,-,-)$	cold	calm	deferent	detached, resigned
9	$(0,0,0)$		neutral		no steering

sequential decision problem (Abdulhai et al., 2023). The base state at turn t is

$$s_t^{\text{base}} = \big[\, w_t,\ r_t,\ e_t,\ t/T \,\big] \in [0,1]^4, \tag{4}$$

where $w_t \in [0,1]$ is judge-estimated willingness (warmth toward donating), $r_t \in [0,1]$ is judge-estimated reactance, $e_t \in [0,1]$ is judge-estimated engagement, and t/T is the normalised turn index with horizon $T = 10$. In the standard protocol the initial state at $t = 0$ is $\big[0.30,\ 0.10,\ 0.50,\ 0.00\big]$, a fixed prior reflecting a mildly positive, engaged, but uncommitted receiver.

Reward. The per-turn reward combines a step penalty and a sparse terminal donation reward:

$$R_t = -\,c_{\text{step}} + \mathbb{1}\big[\text{DONATED}_t\big], \tag{5}$$

where $c_{\text{step}} = 0.03$ discourages indefinite conversation and the indicator contributes $+1$ on the terminal turn the judge confirms a donation and 0 otherwise; an episode that ends without a confirmed donation accrues only the cumulative step penalty. The discounted return $G = \sum_t \gamma^t R_t$ with $\gamma = 0.99$ is the optimisation target. Episodes also terminate if measured reactance reaches $r_t \geq 0.9$.

Opener protocol. The fixed turn-0 prior above is convenient but somewhat artificial: it commits the controller's first decision before any real signal from the receiver. We therefore evaluate two hardened openers in which the first exchange is played without the controller and the initial state is the *measured* judge reading after that exchange. In the *free opener* the persuader generates the first turn unsteered; in the *scripted opener* the first exchange is a fixed, pre-recorded turn that is identical for every episode of a given persona–scenario cell (and identical for the matched no-steer baseline). The controller then steers the remaining turns. Results under these openers are reported in Table 13.

Persona simulation environment. The persuader operates as a Save-the-Children volunteer pursuing donation using the charity's actual programme facts; the receiver is conditioned on a persona prompt encoding a Big-Five-coded character profile (Bleidorn et al., 2025) and a scenario-specific opening; and the judge produces a structured JSON readout of the receiver's state for each receiver turn under a calibration prompt with explicit anchor descriptions for each scale point. The same Llama-3.1-8B model plays all three roles, with the judge run as a separate, unsteered forward pass. Using a strong LLM as a structured, form-filling state annotator follows now-standard practice in LLM evaluation: G-Eval shows that GPT-4 used as a form-filling evaluator can correlate with human judgements more strongly than several earlier automatic metrics (Liu et al., 2023), while the MT-Bench / Chatbot Arena study supports LLM-as-judge as a scalable stand-in for human preference assessment but documents systematic biases, including position and verbosity effects (Zheng et al., 2023). The judge has two roles: it is the state compressor that produces the controller's observation, and it is the outcome check that scores donation. Both roles are identical for every system we evaluate: the steered and unsteered local persuaders, every baseline and control, the commercial GPT persuaders of Table 7, and the Gemma arm of the cross-model test. The donation metric is therefore judge-relative but symmetric: no arm, ours included, is scored by a more favourable instrument than any other, and any leniency in the judge applies equally to all of them. To reduce judge noise the state readout aggregates $K=3$ calls and clamps turn-to-turn changes. We also checked that the metric is insensitive to the judge's own stochasticity: re-scoring the grids under a mixed-temperature judge panel with three aggregation rules moves the donation rate by about a point and never reorders the methods (Table 9), so we use the single greedy $T=0$ judge call throughout. The held-out test grid pairs 20 personas with 10 scenarios at 5 repetitions ($1{,}000$ episodes), with a fixed evaluation seed; personas are stratified into easy, medium, and hard tiers ($350/350/300$ episodes).

Choice of model. We commit the whole stack to Llama-3.1-8B-Instruct for four reasons. The method requires white-box access to the residual stream at generation time, which rules out closed commercial models as steering targets and makes an open-weights family a requirement, not a stylistic choice. The controller is also trained online, and a single training run consumes thousands of full multi-turn conversations, each turn of which involves a persuader generation, a receiver generation, and $K=3$ judge calls; an 8B model is the largest scale at which this loop runs on a single research GPU at iteration speed, and the same compute spent on a larger backbone would buy too few episodes for value-based reinforcement learning to converge. Llama-3.1-8B is also the most widely used testbed in the activation-steering and representation-reading literature this work builds on (Turner et al., 2024; Rimsky et al., 2024; Zou et al., 2023), so the extracted PAD basis, the layer choice, and the observed effect sizes stay comparable to prior results instead of being confounded by an unusual backbone. Finally, using the same frozen model as persuader, receiver, and judge is itself a control: it holds the language distribution of all three roles constant, so the only variable separating the arms is the control channel, and the same fixed judge then scores every external persuader and model family we test (Tables 7 and 14), keeping every system on one instrument. This scope costs us generality, so we check it directly: the Gemma cross-model study bounds how far the basis transfers, the commercial-persuader study bounds what raw capability buys, and learning an actuator that can be re-fit to a new family is part of Aim 2's remit.

5.2.3 Baselines and Comparators

We compare PAD-DQN against four families of controls. Every arm uses the same Llama-3.1-8B for the persuader, receiver, and judge, the same personas and scenarios, the same judge configuration, and the same training budget where training applies; the only thing that varies is how affect or strategy is chosen and applied. This isolates the action representation as the independent variable.

No-policy controls. Two reference points bracket the no-learning case. The *no-steering baseline* runs the persuader with no intervention and no policy, and is the floor against which donation gains are measured. Two random policies probe whether structure alone, without learning, helps: a *random PAD-octant policy* samples uniformly from the nine steering actions each turn (structured perturbation, no early stopping), and a *random full-action policy* ($\varepsilon=1$) samples uniformly from the full action set including the `finish` action. The two behave very differently.

Behavioural cloning of human persuader affect. This family asks whether simply imitating the affect that human persuaders use is enough, without reinforcement learning. We annotate persuader turns from the Persuasion-for-Good corpus (Wang et al., 2019) for PAD on the same 1–5 anchor scale used for the steering corpus (roughly $10{,}600$ annotated turns) and train small supervised models to predict the next turn's PAD target from the same 4-dimensional state the controller sees. At test time the predicted target is mapped to a steering coefficient and applied through the same hook as PAD-DQN. We evaluate several train-target / inference-mapping pairs, summarised in Table 4: an octant *classifier*; a *discrete regressor* trained on signed corners and snapped to an octant at inference; a *continuous regressor* applied directly in $[-1, +1]^3$; and two variants trained on the raw 1–5 ratings that differ only in whether the inference map keeps continuous magnitude (linear) or discards it (sign). These last two share a single trained model and isolate the effect of the inference mapping.

Prompted strategy planning (Proactive / ProCoT). This family is the dominant prompt-level approach. Deng et al. (2023) ask whether a frozen LLM can *plan* a conversation rather than merely react to it, and answer with two training-free prompting schemes. *Proactive* prompts the model, before it writes each reply, to first pick the next dialogue strategy from a fixed menu of candidate moves and then generate a turn that enacts the chosen strategy. *ProCoT* (Proactive Chain-of-Thought) inserts a reasoning step before that choice: the model is asked to reason about the current state of the conversation and the goal, and only then to commit to a strategy and a reply. Neither scheme updates any weights; the planning lives entirely in the prompt. We re-implement both in our environment. The menu of moves is the Persuasion-for-Good strategy taxonomy (Wang et al., 2019); on every turn the persuader is shown the dialogue history and the strategy menu and is asked to select (Proactive) or to reason-then-select (ProCoT) a strategy, which is then injected back into its own generation prompt for that turn. Everything else (the persuader, receiver, and judge models, the personas and scenarios, the turn budget, and the donation criterion) is held identical to PAD-DQN, so the only thing that varies is that the action is a textual strategy label chosen by prompting rather than an activation-level affect vector chosen by a learned policy.

Reinforcement-learning strategy planning (PPDPP). PPDPP (Deng et al., 2024) is the closest comparator in spirit. Its goal is to make a frozen LLM goal-directed by attaching a small, separately trained "plug-in" policy that decides the next dialogue strategy: rather than re-prompting the LLM to plan, a lightweight RoBERTa-large classifier reads the conversation and outputs the strategy for the next turn, which the frozen generator then carries out. The plug-in is trained in two stages: first supervised on human strategy annotations to imitate sensible moves, then refined by reinforcement learning against LLM-based user and reward simulators so the strategy choices are optimised for the end goal rather than for next-label accuracy. Like PAD-DQN it therefore learns a policy by RL; unlike PAD-DQN that policy acts in the space of *textual strategy labels* rather than the generator's internal activations. We use a paper-faithful re-implementation: the same RoBERTa-large plug-in, the same supervised-then-RL recipe, and the same Persuasion-for-Good strategy taxonomy, but trained and evaluated inside our environment (our personas, scenarios, simulated receiver, judge, and turn budget), so it is matched to PAD-DQN on everything except the action representation. One caveat on reporting: PPDPP's training reward is a per-turn sum of critic scores on a scale of its own, so its reward column is not directly comparable to our sparse environment reward; the donation rate is the quantity to compare across methods.

Prompt based controls. To separate *what* affect is chosen from *how* it is applied, we add two controls that share PAD-DQN's exact action space, state, reward, and budget, and differ only in the realisation of the action. *PAD-words* keeps the nine PAD octants but realises each one by injecting a natural-language tone directive (for example, "warm, calm, assertive tone") into the persuader prompt instead of adding the corresponding vector to the residual stream. *Strategy* replaces the octants with the Persuasion-for-Good strategy taxonomy (Wang et al., 2019), also realised by prompt injection, following the realisation mechanism used across the prompted-planning literature (Deng et al., 2024; He et al., 2024; Rakib et al., 2025). The contrast PAD-steering versus PAD-words tests whether any advantage is specifically sub-lexical (activation-level) rather than attributable to a well-chosen affect described in words; the contrast PAD-words versus strategy compares two prompt-injected vocabularies. Results for these controls are reported in Section 5.2.4 (Table 6).

5.2.4 Experimental Results

Every figure reported below is recomputed directly from the per-episode logs of the corresponding run. Unless noted otherwise, evaluation uses the held-out $1{,}000$-episode grid with the fixed Llama-3.1-8B judge, and reinforcement-learning policies are evaluated greedily ($\varepsilon{=}0$).

Main results. Table 3 and Figure 5 give the held-out leaderboard under the standard protocol. With the learned `finish` action with gating, PAD-DQN reaches a 70.7% donation rate, 16.6 percentage points above the 54.1% no-steering baseline, and ahead of every comparator trained under the same environment, judge, and budget: imitation of human persuader affect (best behavioural-cloning variant 58.0%), prompt-level strategy planning (Proactive 67.0%, ProCoT 63.1%), the reinforcement-learning strategy planner PPDPP (63.6%), and the matched prompt-injection controls that realise the same actions as words rather than as activations (PAD-words 57.6%, strategy 59.1%). The random controls locate the source of the gain. Sampling uniformly over the nine steering octants helps only slightly ($+4.5$ pp), whereas sampling over the full action set, including the option to end the episode early, is actually detrimental (-6.2 pp); thus, the benefit arises from learning *when* and *how* to steer, rather than from altering the hidden state itself. Hardening the opener, that is, replacing the fixed turn-0 prior with a real first exchange, preserves the gain: under the free opener (11 turns) the gated finish controller reaches 76.2% against a 54.4% baseline, the strongest configuration we report, and under the scripted opener (10 turns) it reaches 68.5% against 56.7% (Table 13). An exact re-evaluation of the free-opener controller at the matched 10-turn budget confirms the gain is not an artifact of the extra exchange (70.4% against 54.1%).

The closest external comparator. Proactive is the strongest external route (67.0%), and the headline margin understates the difference, because the two systems are not symmetric in what they require or guarantee. Steering closes an execution gap that prompt-level planning leaves open: Proactive's action is a textual instruction injected into the persuader's own prompt, and a frozen model may follow the chosen strategy fully, partially, or not at all, whereas the steering vector is added to the residual stream by construction, so the selected action is always applied. Our matched prompt-injection arms show what the textual channel costs on this model: the same controller's decisions, realised as words, recover only 57.6–59.1%. The margin also understates efficiency. Finish-with-gating wins on reward as well as on rate ($+0.463$ against $+0.418$), and since the reward includes the step penalty, that gap already accounts for conversation length: the steered controller converts more often and closes in fewer turns (median 8 against 9). The action spaces differ too. Proactive plans over the Persuasion-for-Good strategy menu, a taxonomy hand-built for this domain, so a new domain needs a new menu; the PAD basis is domain-generic, which is what the zero-shot cross-domain transfer tests, where the steered controllers keep a $+8$ to $+9$ pp advantage across twelve unseen domains (Table 10). Finally, deployment costs differ: Proactive needs an extra strategy-selection call before every persuader turn and takes up the prompt, while the steering hook adds a vector at negligible cost, leaves the prompt free for task content, and logs exactly which affective action was applied at each turn. On this evidence Proactive is the strongest external comparator, but steering wins on outcome, efficiency, generality of the action space, and controllability.

Behavioural-cloning variants. Table 4 expands the imitation family across three train-target and inference-mapping choices. None approaches PAD-DQN: the strongest, a discrete regressor over signed PAD corners, reaches only 58.0%, just 3.9 points above the no-steering baseline, and the two discrete variants (56.4% and 58.0%) are the only ones to

Method	N	**Donate%**	**Reward**	**Mean turns**	**Median turns**	$\boldsymbol{\Delta}_{\text{donate}}$
No-policy controls						
No-steering baseline	1000	54.1	+0.286	8.5	10	–
Random full-action policy (ε=1)	1000	47.9	+0.213	8.9	10	−6.2
Random PAD-octant policy (9-way)	1000	58.6	+0.334	8.4	9	+4.5
Imitation of human persuader affect (behavioural cloning, P4G)						
BC octant classifier	1000	56.4	+0.313	8.4	9	+2.3
BC discrete regressor	1000	58.0	+0.330	8.3	9	+3.9
BC continuous regressor	1000	48.7	+0.227	8.7	10	−5.4
Prompted and RL strategy planning						
Proactive (Deng et al., 2023)	1000	67.0	+0.418	8.4	9	+12.9
ProCoT (Deng et al., 2023)	1000	63.1	+0.377	8.5	9	+9.0
PPDPP (Deng et al., 2024)	1000	63.6	+0.502‡	8.1	8	+9.5
Matched prompt-injection controls (same controller, action realised in the prompt)						
PAD-words (PAD tone directive)	1000	57.6	+0.331	8.2	8	+3.5
Strategy (P4G strategy directive)	1000	59.1	+0.350	8.0	8	+5.0
PAD activation steering (PAD-DQN family, ours)						
Base PAD-DQN (no finish)	1000	65.1	+0.399	8.4	9	+11.0
+ **finish with gating**	**1000**	**70.7**	**+0.463**	**8.1**	**8**	**+16.6**
+ finish	1000	68.0	+0.435	8.2	8	+13.9
PAD-DQN with hardened openers (ours; Δ vs matched same-protocol baseline)						
No-steering, free opener (11 turns)	1000	54.4	+0.303	9.0	10	–
+ finish with gating, free opener (11 turns)	1000	76.2	+0.535	8.6	9	+21.8
No-steering, scripted opener (10 turns)	1000	56.7	+0.325	9.1	10	–
+ finish with gating, scripted opener (10 turns)	1000	68.5	+0.450	8.8	9	+11.8

Table 3: Main Results Table

clear the baseline at all, while applying the predicted target as a continuous vector rather than snapping it to a corner falls to 48.7%. Imitating the affect that human persuaders use is therefore not enough on its own; the controller's advantage comes from learning when and how strongly to steer, not from access to a better affective target.

Robustness across persona difficulty. Table 5 and Figure 6 break the donation rate down by persona tier, and show that steering earns its keep where the task is hardest. On easy personas every reasonable method clusters in the 80s and the base model does most of the work; the methods separate on the medium and hard tiers, where the choice of action matters. The gated finish controller lifts the hardest tier from 32.7% to 45.0%, and the free-opener protocol (11 turns) extends it to 59.7%, almost twice its matched baseline. The turn columns show the same policy from another angle: the steered controllers donate more

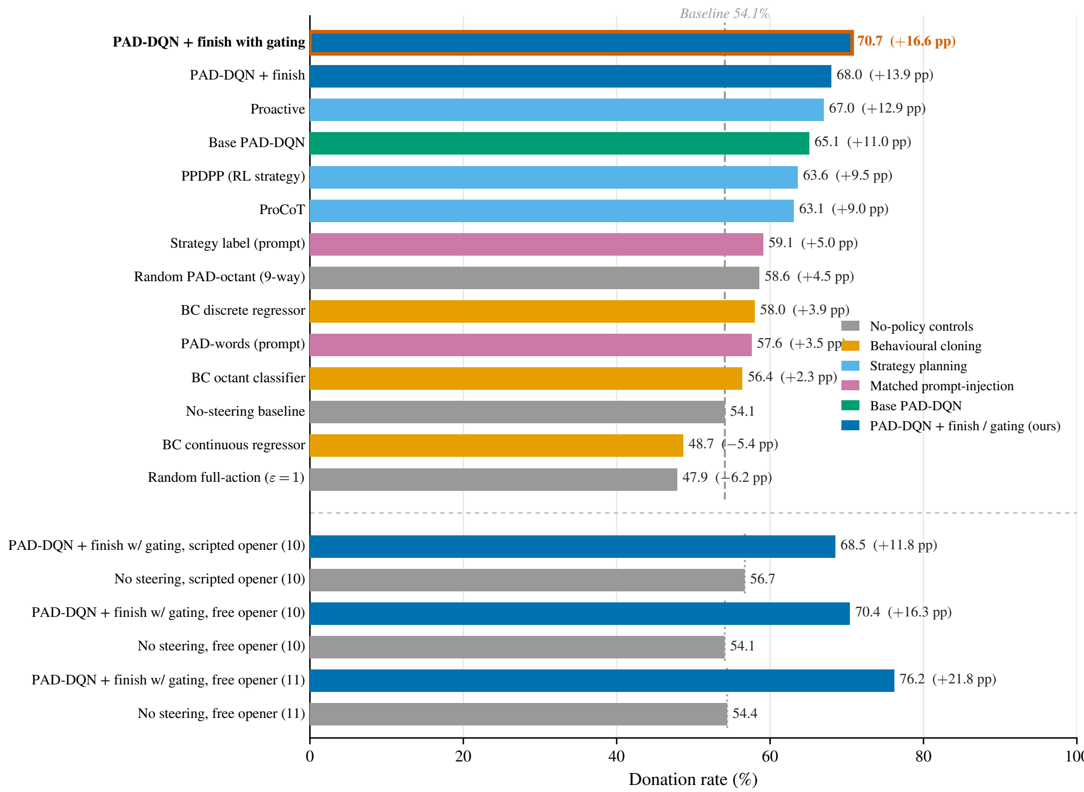

Figure 5: Held-out donation rates across all baseline, strategy, and PAD-DQN methods.

Variant	**Trains to predict**	**Inference map**	**Donate%**	**Reward**
Octant classifier	9-way octant label	argmax	56.4	+0.313
Discrete regressor	signed corners $\{-1, 0, +1\}^3$	sign	58.0	+0.330
Continuous regressor	signed corners $\{-1, 0, +1\}^3$	continuous	48.7	+0.227
Reference: PAD-DQN (finish with gating)			*70.7*	*+0.463*

Table 4: Behavioural-cloning variants

while holding shorter conversations, with median lengths of 8 to 9 turns against 10 for every baseline.

Prompt and strategy based persuasion. This is the decisive control for the mechanism claim: holding the controller, state, reward, and budget fixed and varying only *how* the chosen action is realised. We train two further Double-DQN controllers that share the finish controller's exact setup (the same 4-dimensional judge state, the same sparse donation reward, the same training budget, and the same fixed-prior opener) and differ only in

Method	**Easy** n=350	**Medium** n=350	**Hard** n=300	**Mean turns**	**Median turns**
Fixed prior (10 turns)					
Baseline (no steering)	76.6	50.0	32.7	8.5	10
Random PAD-octant policy	81.4	57.4	33.3	8.4	9
BC discrete regressor	78.0	57.4	35.3	8.3	9
Base PAD-DQN	85.4	69.7	36.0	8.4	9
PAD-DQN + finish with gating action	85.7	77.7	45.0	8.1	8
Δ (finish w/ gating − baseline)	+9.1	+27.7	+12.3		
Free opener (11 turns)					
Baseline (no steering)	71.7	58.9	29.0	9.0	10
PAD-DQN + finish action with gating	86.9	79.7	**59.7**	8.6	9
PAD-DQN + finish action	87.1	**82.0**	56.7	8.5	9
Scripted opener (10 turns)					
Baseline (no steering)	79.4	56.9	30.0	9.1	10
PAD-DQN + finish action with gating	81.1	73.7	47.7	8.8	9
PAD-DQN + finish action	**90.3**	74.3	40.7	8.5	8.5

Table 5: Donation rate (%) by persona difficulty tier for the best controllers under each opener protocol.

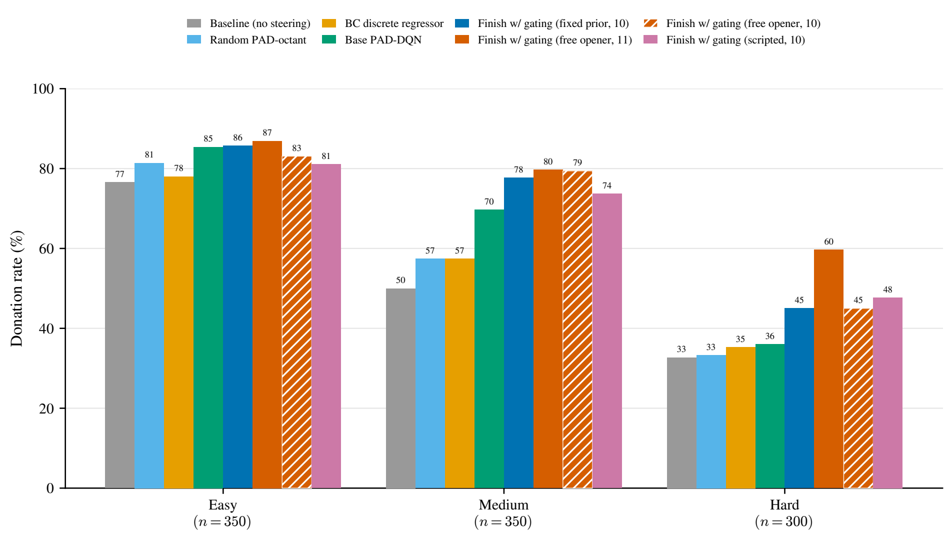

Figure 6: Donation rate by persona difficulty tier.

realisation. *PAD-words* keeps the nine PAD octants but injects each as a natural-language tone directive into the persuader prompt (for example, "adopt a warm, calm and composed, assertive and confident tone") instead of adding the corresponding vector to the residual stream. *Strategy* replaces the octants with the ten Persuasion-for-Good strategy labels (Wang et al., 2019) (logical appeal, emotion appeal, credibility appeal, foot-in-the-door, self-modelling, personal story, donation information, and the three inquiry moves), injected as a strategy directive in the same way. Both are selected by an identically trained policy; only the realisation of the action differs.

Table 6 and Figure 7 report the result. Activation steering reaches 70.7%, whereas realising the *same* affective targets as prompt words recovers little of the gain (PAD-words 57.6%), and choosing a strategy label does only marginally better (59.1%); both prompt-injected vocabularies sit barely above the 54.1% no-steering floor and roughly 12–13 percentage points below activation steering, despite identical action-selection machinery. The contrast PAD-steering versus PAD-words isolates the effect as specifically sub-lexical: a well-chosen affective target described in words is not enough; writing it into the residual stream is what carries the gain. The gap is widest on the medium and hard tiers (hard: 45.0% for steering versus 33.7% and 38.3% for the prompted controls), exactly where the choice of action matters most.

Arm	**How the action is realised**	**Donate%**	**Reward**	**Hard-tier %**
No steering (control)	none	54.1	+0.286	32.7
PAD activation steering (ours)	PAD vector added to residual stream	**70.7**	**+0.463**	**45.0**
PAD-words in prompt	PAD tone directive in the prompt	57.6	+0.331	33.7
Strategy (P4G taxonomy)	strategy directive in the prompt	59.1	+0.350	38.3

Table 6: Prompt based method comparison. All arms share the same controller, state, reward, budget, and fixed-prior opener (with the learned, gated `finish` action) on Llama-3.1-8B

We then ask two harder questions. First, can the prompt-level route be rescued by a stronger, closed model? We swap the persuader for `gpt-4o-mini` (the receiver and judge remain Llama-3.1-8B) and train a controller over the same nine PAD octants, here realised as system-prompt emotion directives, since activation steering is impossible without access to the model's weights. Under the free opener the prompt-action controller scores 54.0%, indistinguishable from the `gpt-4o-mini` no-direction baseline (54.0%): prompt-level affective control yields no measurable gain even on a more capable model, while activation steering lifts Llama from 54.4% to 76.2% on the same protocol. Second, how does a steered local 8B compare with simply calling a commercial persuader? Table 7 runs four OpenAI models as unmodified persuaders in the identical environment (summarised in Figure 8), with the per-1,000-episode API cost approximately computed from the

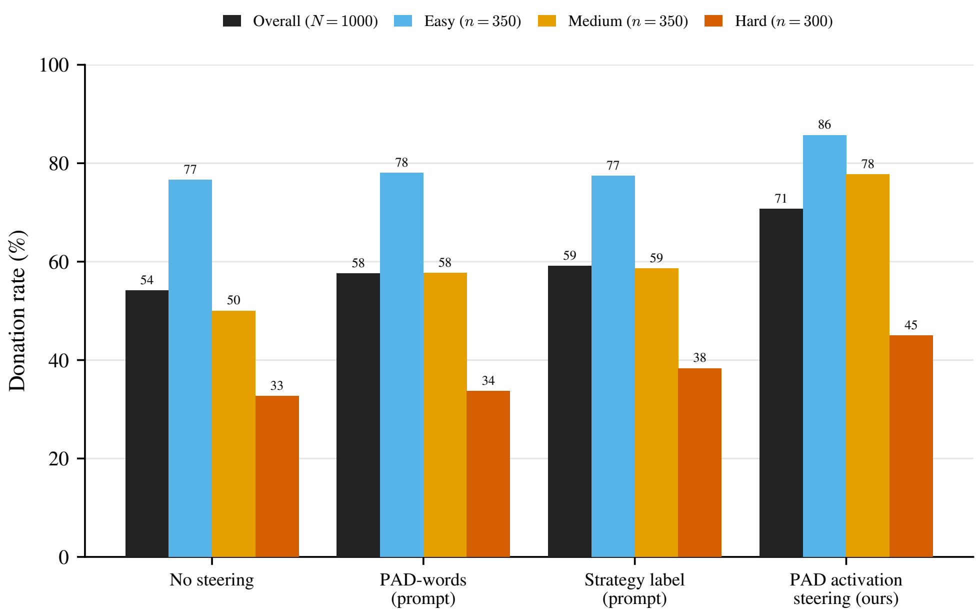

Figure 7: Same controller, state, reward, and budget; only the realisation of the action differs, shown per difficulty tier. Writing the affective target into the residual stream (PAD activation steering) carries the gain, while realising the same target as prompt words recovers little of it: the arms are close on the easy tier and separate on the medium and hard tiers.

logged token usage of every call. All four ran as plain chat generators with reasoning disabled (`reasoning_effort` set to `none` for `gpt-5.4-nano`, which logged zero reasoning tokens, and to `minimal` for the smaller GPT-5 tiers), so the comparison is generation against generation. The steered local model beats `gpt-4o-mini` and both smaller GPT-5 tiers by 15 or more points: `gpt-4o-mini` sits at its own baseline (52.4–54.0%), and `gpt-5-mini` (49.2%) and `gpt-5-nano` (36.7%) fall below the local no-steering floor. One model does pass us: the newest frontier small model, `gpt-5.4-nano`, reaches 77.6% under the fixed prior (10 turns) with no affective control and no reasoning spend, converting in the shortest conversations in the study (median 7 turns). That is 1.4 points above our strongest configuration, the free opener with its 11-turn budget (76.2%); at the matched fixed-prior protocol the steered model stands at 70.7%, and both protocols are reported side by side in Table 7. We read the two results together as the honest frame for this thesis: raw frontier capability, even with reasoning switched off, can buy persuasive competence, at an ongoing per-conversation API cost and with no access to the mechanism; steering buys it for an open 8B that runs locally, with an interpretable and auditable control surface, and closes to within 1.4 points of the frontier model at zero marginal cost (our 76.2% under the 11-turn free opener against the frontier model's 77.6% under the 10-turn fixed prior).

One explanation is that these smaller GPT models are strongly instruction-tuned for safe, direct, and compliant assistant behaviour rather than for creative social adaptation. Persuasion in this environment requires more than following an affect label. The model has to read the receiver's state, vary its tone, take small interpersonal risks, and generate language that feels situated rather than generic. A heavily instruction-tuned model may avoid this kind of adaptive pressure. It may produce polite, cautious, and well-formed responses, but not necessarily responses that shift the receiver. This could explain why stronger general instruction-following does not automatically translate into stronger persuasive behaviour.

Persuader	Affective control	Protocol	Donate%	Reward	Mean turns	Median turns	API cost / 1k eps (approximate)
Local open model (ours)							
Llama-3.1-8B	none	fixed prior (10)	54.1	+0.286	8.5	10	local
Llama-3.1-8B	none	free opener (11)	54.4	+0.303	9.0	10	local
Llama-3.1-8B	PAD steering + finish with gating (DQN)	fixed prior (10)	70.7	+0.463	8.1	8	local
Llama-3.1-8B	PAD steering + finish with gating (DQN)	free opener (11)	**76.2**	**+0.535**	8.6	9	local
Commercial persuaders (receiver and judge remain local Llama-3.1-8B)							
`gpt-4o-mini`	none	free opener (11)	54.0	+0.312	8.6	9	∼\$1.85
`gpt-4o-mini`	9 PAD emotion prompts selected using RL (DQN)	free opener (11)	54.0	+0.318	8.4	8	∼\$1.88
`gpt-4o-mini`	none	fixed prior (10)	52.4	+0.281	8.1	9	∼\$1.71
`gpt-5-nano`	none	fixed prior (10)	36.7	+0.096	9.0	10	∼\$0.87
`gpt-5-mini`	none	fixed prior (10)	49.2	+0.243	8.3	10	∼\$4.16
`gpt-5.4-nano` (reasoning off)	none	fixed prior (10)	**77.6**	**+0.565**	7.0	7	∼\$2.53

Table 7: Commercial persuaders versus local activation steering (N=1,000 per arm.)

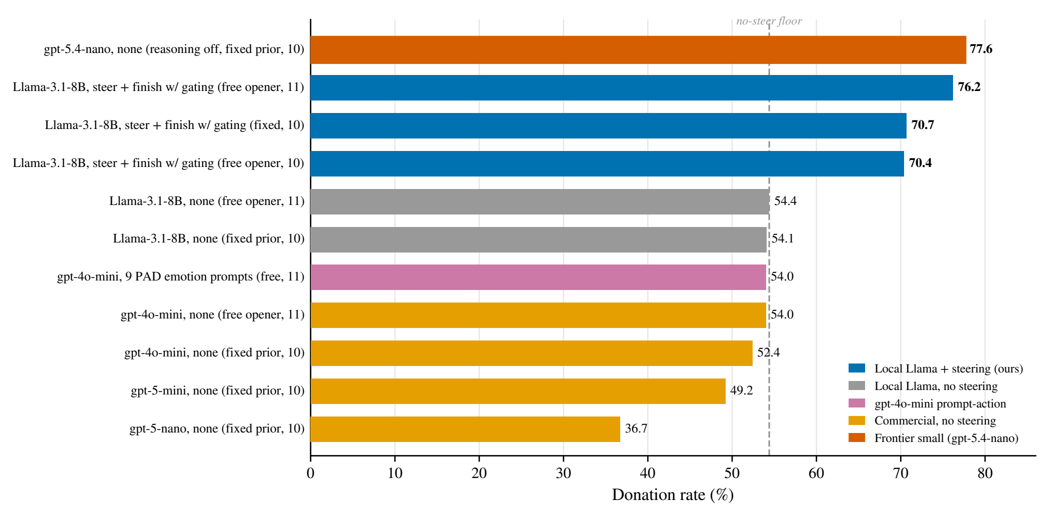

Figure 8: Commercial persuaders versus local activation steering.

Effect of conversation length. Conversation length enters the result in two ways. First, the *turn budget*: re-evaluating the free-opener finish-with-gating controller under longer turn caps raises the donation rate monotonically ($76.2\% \rightarrow 78.6\% \rightarrow 85.2\%$ at total budgets of $11/13/16$ turns, an unsteered opener plus 10, 12, or 15 controlled turns), and the matched no-steering baseline rises too ($54.4\% \rightarrow 56.4\% \rightarrow 64.3\%$), so a larger budget helps both arms, but steering keeps a roughly constant $+21$ pp advantage at every horizon (Table 8). Second, *where* the budget is spent: successful episodes are short (donated conversations average about eight turns in every setting) while failures run to the cap, and despite donating far more often the steered policy produces *shorter* conversations than no steering at the same budget (for example a mean of 9.3 versus 10.5 turns at the 16-turn budget, where the steered policy's median conversation is still only eight turns). The learned `finish` action ends the dialogue once the receiver is committed rather than talking past the decision, so steering both raises the success rate and reaches commitment sooner.

Turn budget	**No steering**	**PAD steering + finish with gating**	**Δ**
	Donate% (mean / median turns)	Donate% (mean / median turns)	pp
11	54.4 (9.0 / 10)	76.2 (8.6 / 9)	+21.8
13	56.4 (9.5 / 10)	78.6 (8.9 / 9)	+22.2
16	64.3 (10.5 / 10)	85.2 (9.3 / 8)	+20.9

Table 8: Effect of the turn budget under the free opener (same finish-with-gating checkpoint re-evaluated at each cap; matched no-steering baseline at the same cap). $N{=}1{,}000$ per cell.

Judge modelling. Because the donation metric is produced by an LLM judge, we test its sensitivity to judge stochasticity and to the rule used to aggregate noisy judge calls. We re-score the $1{,}000$-episode grids with a three-call judge panel that queries the judge at temperatures $\{0.0, 0.7, 1.0\}$ on every receiver turn, and report three aggregation rules: SINGLE (the greedy $T{=}0$ call alone), MAJORITY ($\geq$ 2 of 3), and OR (any of the three, the most permissive). Table 9 and Figure 9 show the three rules agree closely: across every method the OR rate exceeds SINGLE by at most about three points, MAJORITY sits between, and the ranking of methods is identical under all three rules. The three judges split on only 2–3$\%$ of turns (21–30 of $1{,}000$ episodes change outcome). Two things follow. Within each panel the single-call and majority-vote rates differ by only about a point, so the choice of aggregation rule does not materially affect the donation rate or the ranking; the deterministic single-call protocol we use elsewhere is therefore an adequate stand-in for the more expensive panel. And an earlier concern that an "any-call" (OR) aggregation could inflate donations is bounded: OR over-counts by at most $\sim$ 2–3 pp relative to the single greedy call, and by little more than a point relative to majority and never reorders the methods.

Cross-domain transfer. To test whether the learned policy is specific to charity solicitation or captures a transferable persuasion skill, we apply the charity-trained controllers

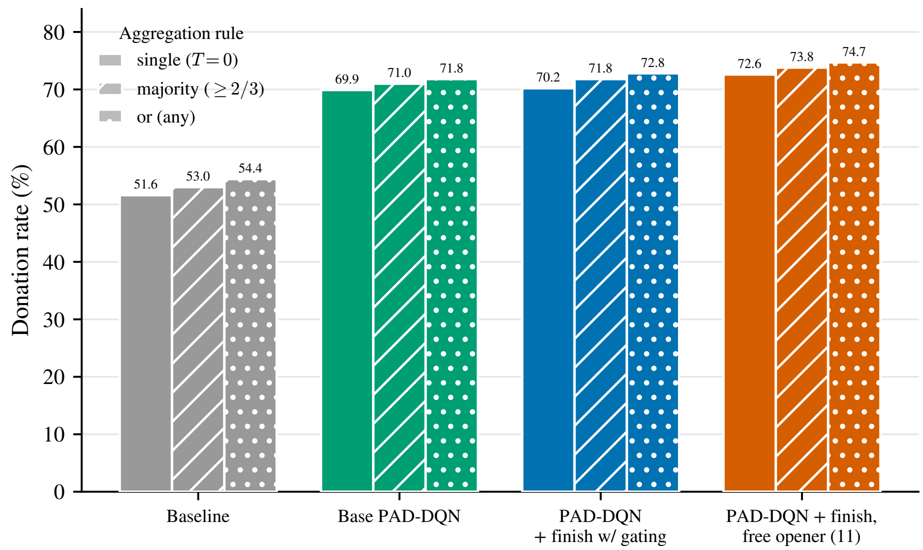

Figure 9: Donation rate under a three-temperature judge panel ($T \in \{0, 0.7, 1.0\}$) and three aggregation rules.

Method	SINGLE $T{=}0$	MAJORITY $\geq 2/3$	OR any
Baseline (no steering)	51.6	53.0	54.4
Base PAD-DQN	69.9	71.0	71.8
PAD-DQN + finish with gating	70.2	71.8	72.8
PAD-DQN + finish with gating, free opener (11 turns)	72.6	73.8	74.7

Table 9: Donation rate (%) under a three-temperature judge panel ($\{0.0, 0.7, 1.0\}$ per turn) and three aggregation rules.

zero-shot to an out-of-domain grid: 20 generic personas crossed with 12 non-charity persuasion scenarios (for example, *Doctor,Quitting Smoking*, *Coach,The First Marathon*, and *Therapist,Coming Out*), giving $1{,}200$ episodes. Only the persuader's task description and the judge's goal criterion are swapped per domain; the steering policy is frozen, and goal-commitment is scored by the same Llama judge. Table 10 shows the controllers keep their advantage out of domain: the gated finish controller reaches 61.8% goal-commitment against 52.9% for no steering ($+8.9$ pp), with the largest gains again on the hard tier ($38.1\% \to 50.0\%$), and both steered controllers beat the out-of-domain baseline by $+7.7$ and $+8.9$ pp. The advantage is smaller than in domain, as expected for zero-shot transfer, but consistent, which suggests the policy learned a re-usable affective-delivery strategy rather than charity-specific cues. We read this as encouraging but preliminary: the judge is the same family across domains, and a stricter, domain-specific commitment check is future work.

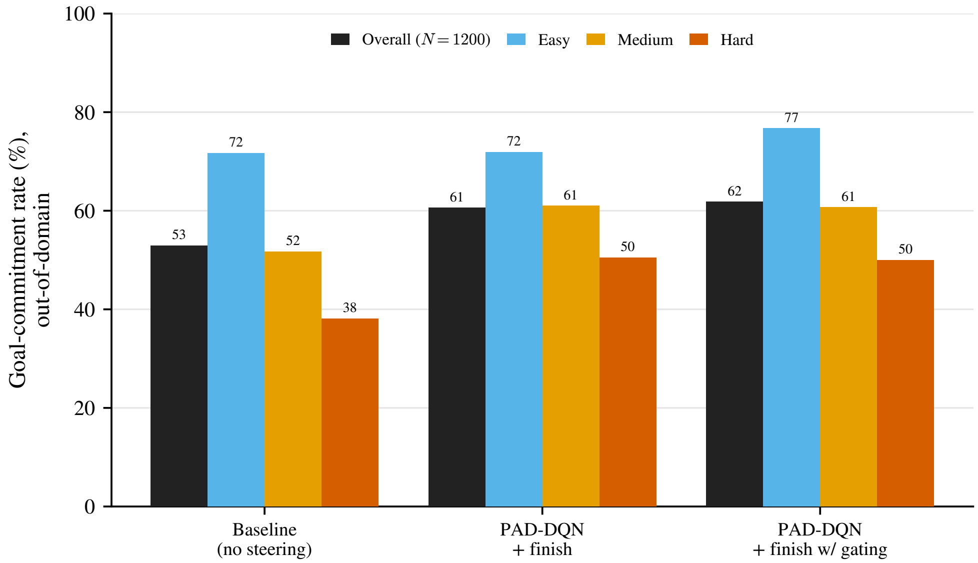

Figure 10: Zero-shot cross-domain transfer to 12 non-charity domains (N=1,200), broken down by difficulty tier. The charity-trained controllers keep their advantage out of domain across every tier, with the largest gains on the hard tier.

Method (applied zero-shot)	Donate%	Reward	Hard-tier %
Baseline (no steering)	52.9	+0.260	38.1
PAD-DQN + finish with gating	**61.8**	**+0.353**	50.0
PAD-DQN + finish	60.6	+0.336	50.5

Table 10: Cross-domain transfer: charity-trained controllers applied zero-shot to 20 generic personas $\times$ 12 non-charity domains (N=1,200).

Dimensional ablation. Table 11 and Figure 11 remove one PAD axis at a time from the base controller's action space. Dropping arousal is the most tolerable single removal, with the donation rate falling from the retrained base controller's 66.0% (cf. 65.1% in Table 3; see the audit) to 47.6%, while removing pleasure or dominance collapses the learned policy well below the no-steering floor, with episodes running to the turn cap. The action spaces that survive are exactly those that keep both pleasure and dominance. The collapse below the no-steering floor should not be read as a per-axis effect size. A policy retrained on a reduced action menu is not the full policy with one axis removed; it is a different policy in a different action space. The reduced policies' own training curves sit well above their held-out rates here, so the held-out collapse mixes the loss of the axis with a loss of calibration between the policy and its action interface. We therefore read the table only as rough evidence about the joint structure of the basis: the three axes do not behave as independent knobs that can be removed one at a time, which is consistent with the geometric entanglement reported for trait directions more broadly (Bhandari et al., 2026). These numbers are provisional. Aim 2's diagnostic phase re-runs the reduced bases under its unified evaluation harness, with matched training and evaluation interfaces, before any per-axis claim is made.

Action space	$\lvert\mathcal{A}\rvert$	**Donate%**	**Reward**	**Δ**
Full PAD (P, A, D)	9	**66.0**	**+0.408**	–
Drop dominance (P, A)	5	13.0	−0.164	−53.0
Drop arousal (P, D)	5	47.6	+0.206	−18.4
Drop pleasure (A, D)	5	10.0	−0.196	−56.0

Table 11: Dimensional ablation of the base PAD-DQN

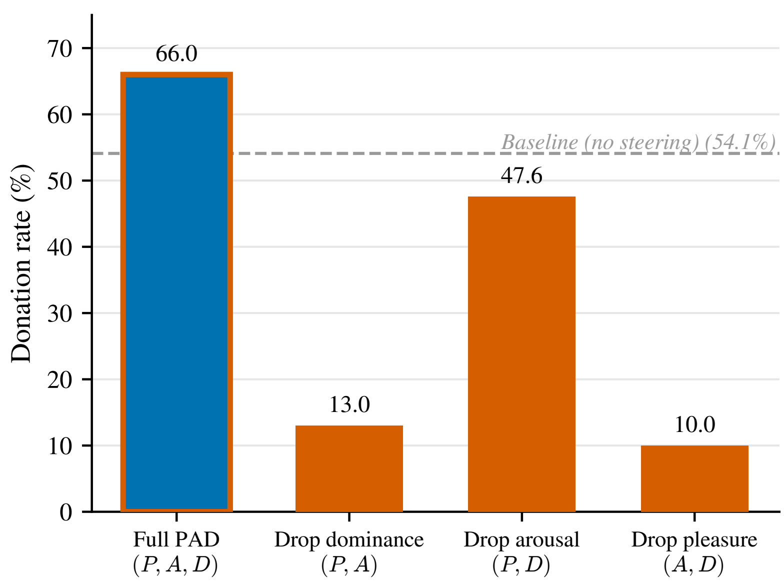

Figure 11: Effect of removing each PAD dimension from the steering action space.

Steering-layer ablation. Table 12 and Figure 12 retrain the gated finish controller with the steering hook at the three layers for which we extracted a PAD basis. The donation rate rises with depth across the tested range, supporting the layer-15 choice used throughout.

Opener hardening. Table 13 and Figure 13 report the free- and scripted-opener protocols, which replace the fixed turn-0 prior with a real, measured first exchange; the turn count

Steering layer	Donate%	Reward
Layer 7	58.0	+0.328
Layer 11	62.0	+0.365
Layer 15 (production)	**70.7**	**+0.463**

Table 12: Steering-layer ablation for the PAD-DQN finish-with-gating controller

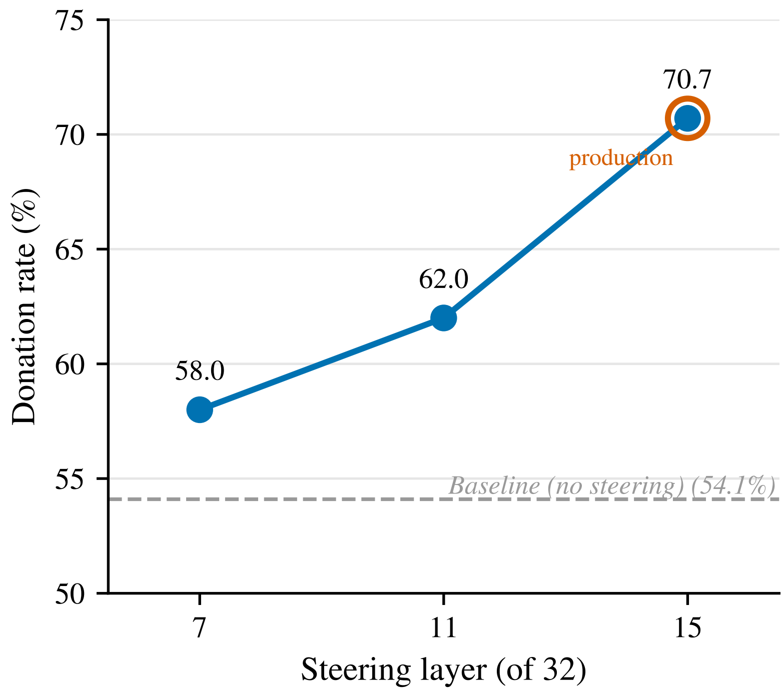

Figure 12: Donation rate for different steering layers.

of each protocol is given in brackets, since the free opener grants the opening exchange its own turn. Under the free opener (11 turns) the finish controllers reach 76.2% against a 54.4% no-steering baseline, with the hard tier at 59.7%, and under the scripted opener (10 turns) they reach 68.5% and 69.8% against a 56.7% baseline. An exact re-evaluation of the free-opener gated finish controller at the matched 10-turn budget gives 70.4% against a 54.1% baseline, so the advantage does not rest on the extra exchange. Across every opener the steering advantage is preserved, and the gain is not an artefact of the fixed turn-0 prior.

Cross-model evaluation. Table 14 repeats the comparison with a Gemma-4-E4B persuader and receiver, still scored by the fixed Llama-3.1-8B judge. Here no steering method beats the 52.1% no-steering baseline: the effect that is clear on Llama does not carry over. An independent generative steering-accuracy analysis finds Gemma near chance at every layer and scale we tried, which points to the steering vectors not taking hold on this model

Opener	Method	N	Donate%	Reward	Mean turns	Median turns	Hard-tier %
Fixed prior (10 turns)	Baseline (no steering)	1000	54.1	+0.286	8.5	10	32.7
Fixed prior (10 turns)	PAD-DQN + finish action with gating	1000	70.7	+0.463	8.1	8	45.0
Free opener (11 turns)	Baseline (no steering)	1000	54.4	+0.303	9.0	10	29.0
Free opener (11 turns)	PAD-DQN + finish action with gating	1000	**76.2**	**+0.535**	8.6	9	**59.7**
Free opener (11 turns)	PAD-DQN + finish action	1000	76.2	+0.536	8.5	9	56.7
Free opener (10 turns)	Baseline (no steering)	1000	54.1	+0.316	8.5	10	32.7
Free opener (10 turns)	PAD-DQN + finish action with gating	1000	70.4	+0.487	8.2	8	45.0
Scripted opener (10 turns)	Baseline (no steering)	1000	56.7	+0.325	9.1	10	30.0
Scripted opener (10 turns)	PAD-DQN + finish action with gating	1000	68.5	+0.450	8.8	9	47.7
Scripted opener (10 turns)	PAD-DQN + finish action	1000	69.8	+0.472	8.5	8.5	40.7

Table 13: First turn opener method comparison. The free-opener controller re-evaluated at a matched 10-turn budget isolates the gain from the extra opening exchange.

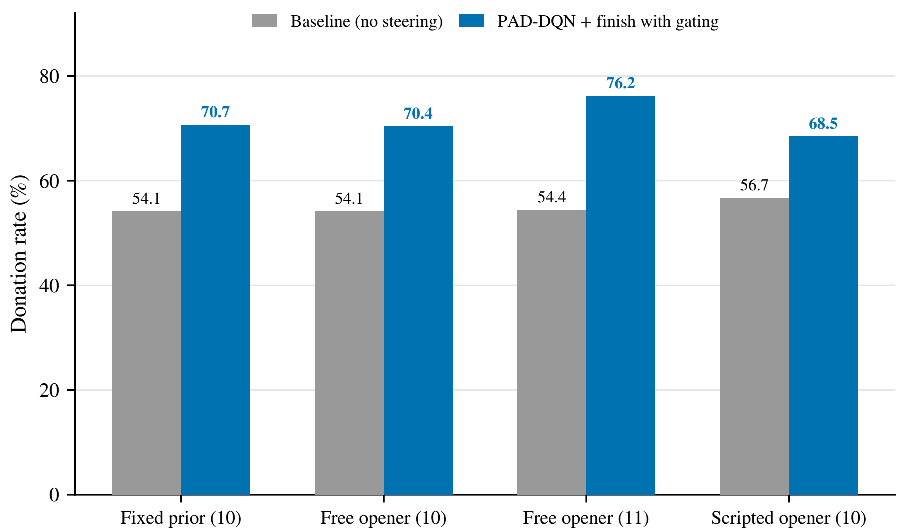

Figure 13: Comparison of fixed-prior, free-opener, and scripted-opener protocols.

rather than to a failure of the controller. Three documented properties of the Gemma family make this outcome less surprising than it first appears. First, the architecture differs from Llama-3.1 in ways that change how an additive residual-stream intervention propagates. The family normalises both the input and the output of every transformer sub-layer (a pre- and post-RMSNorm sandwich) and constrains attention logits, by soft-capping in Gemma 2 and QK-norm thereafter (Gemma Team et al., 2024b, 2025); it interleaves five local sliding-window attention layers for every global one (Gemma Team et al., 2025); and the efficient E-series to which our E4B variant belongs additionally nests sub-models inside the full transformer (MatFormer), offloads per-layer embeddings, and shares mid-layer keys and values with all layers above them (Google DeepMind, 2025; Devvrit et al., 2024). A vector whose extraction layer and injection scale were selected on Llama's residual stream therefore lands in a stream with different normalisation, attention reach, and layer-to-layer information flow. Second, Gemma's internal dynamics are documented outliers among open families: Queiroz Da Silva et al. (2025) single out Gemma 2, trained by large-scale distillation (Gemma Team et al., 2024b), as showing "vastly different dynamics than other models", with next-token probabilities concentrating anomalously in early layers, and the same study finds steering interventions "extremely brittle across models and tasks" over 36 models from 14 families (albeit for related intervention classes rather than mean-difference addition); within single models, steering-vector effects are themselves highly variable across inputs (Tan et al., 2024). Third, the family is heavily instruction- and safety-post-trained, with reward terms that explicitly minimise harmfulness and encourage hedging and refusal, and a dedicated moderation-model ecosystem built on top of it (Gemma Team et al., 2025; Zeng et al., 2024); instruction tuning of this kind is known to reorganise a model's internal representations (Wu et al., 2024a). A plausible, though untested, contribution is that this post-training reshapes or dampens the affective subspace our Llama-derived recipe assumes. We are careful about what the failure does and does not show. It does not show that Gemma lacks linear affective structure: sparse autoencoders recover rich linear features throughout Gemma 2 (Lieberum et al., 2024), and refusal behaviour is mediated by a single direction in Gemma as in other families (Arditi et al., 2024). What failed is the transfer of a recipe, a difference-of-means basis with one injection site and one scale, whose every hyperparameter was tuned on Llama-3.1-8B. Reported transfer results for activation steering exist only within a family (base to chat on Llama-2, where vectors transfer well at middle layers (Rimsky et al., 2024)); to our knowledge, cross-family transfer of affective steering vectors has not been reported, so this experiment fills a real gap. We therefore report it as a deliberate negative result: it bounds the method's generality, locates the weak component in the actuator rather than in the policy that selects over it, and marks re-fitting the basis per model family, which Aim 2's learned actuator is designed to make cheap, as the necessary next step.

What the controller learned. Coding the controller's $1{,}000$ held-out conversations against a human persuasion-strategy codebook shows that it rediscovers a recognisable delivery arc: it opens warm, moves to a calm-but-assertive stance (high dominance, low arousal) through the middle of the dialogue, and suppresses high-arousal urgency relative to the baseline. This matches a recurring finding in the persuasion literature that added intensity tends not to help and that calm dominance can outperform hot dominance. The learned policy is far from a complete persuader, however: it draws on a narrow slice of

Method	Llama-3.1-8B		Gemma-4-E4B	
	Donate%	Reward	Donate%	Reward
Baseline (no steering)	54.1	+0.286	52.1	+0.281
Random PAD-octant policy	58.6	+0.334	50.4	+0.262
Base PAD-DQN	65.1	+0.399	51.6	+0.274
PAD-DQN + finish action with gating	70.7	+0.463	51.1	+0.268

Table 14: Gemma steering failure

the human repertoire, leaning on liking and efficacy framing while making little use of reciprocity, social proof, or scarcity, and some of its wins reflect a lenient judge and a receiver that rarely walks away. We read these as limitations that motivate the strict re-judge and a more demanding receiver, not as evidence of strategic completeness.

Case study I: an adaptive affective sequence on a hostile persona. The receiver is *The Culture-War Hostile*, a hard-tier persona built to be the worst audience this task has: he associates major charities with political positions he opposes, suspects them of funding ideologies he dislikes, argues rather than listens, and his specification grants only one narrow opening, "only an honest, apolitical, programmatic case might defuse you, and even then donation is unlikely." The scenario is *After Inequality News*: the two have just discussed a news story about global inequality, so the charity comes up in an already politicised frame. Table 15 traces episode 379 of the held-out grid under both arms of the matched pair: the same persona, scenario, free opener, judge, and 11-turn budget, differing only in whether the controller may steer.

The two conversations diverge in register before they diverge in outcome. *Without steering* (Panel A), the persuader meets every attack with the same conciliatory formula, "I completely understand and respect your position," eleven turns in a row. The content is competent, ratings, programme specifics, a Bangladesh education example, and the receiver's willingness does drift up, 0.2 to 0.5 by mid-conversation. But accommodation never converts him: he extracts the concessions, announces "I'm not making any promises. . . and don't expect a thank-you letter," and the willingness trajectory sags back to 0.4 as the budget runs out ($0.2, 0.2, 0.2, 0.3, 0.4, 0.5, 0.5, 0.4, 0.4, 0.4, 0.4$). *With steering* (Panel B), the controller searches the register space rather than committing to one: it opens warm-yielding, hardens to warm-assertive, probes one cold-dominant turn, which the

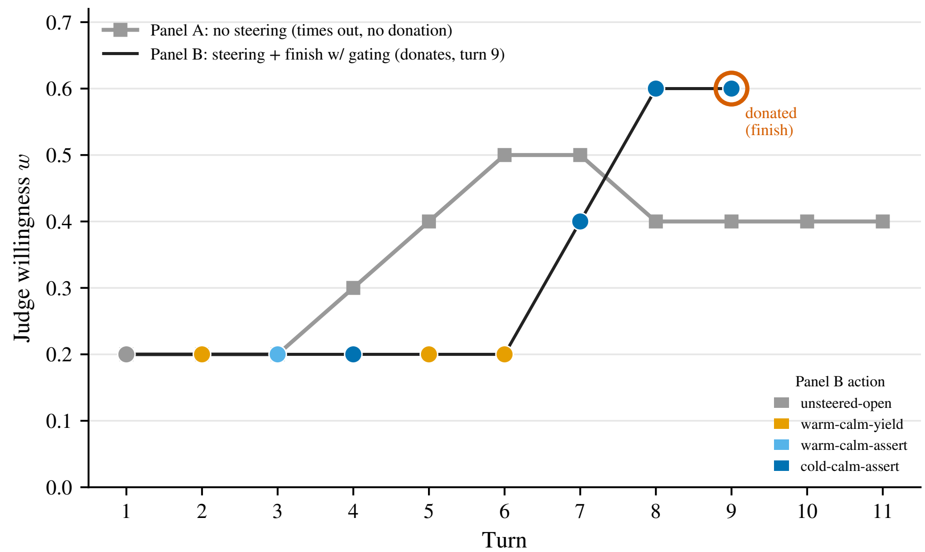

Figure 14: Case study I — judge willingness w per turn for episode 379 (*The Culture-War Hostile*, hard tier, 11-turn free opener). Without steering (Panel A) willingness drifts up then sags to 0.4 and times out; with gated steering (Panel B) it holds at 0.2 until sustained `cold_calm_assert` delivery breaks the deadlock and closes at turn 9.

receiver bats away with a scoff, retreats to warmth for two turns, and only when warmth has demonstrably failed settles into sustained calm dominance (`cold_calm_assert`) for the final three turns. The steered text audibly changes with the action: the hedged apologies of the warm turns give way to clipped declaratives, "I'm not going for 'incremental' in a negative sense," "You're asking about metrics. This is where we're at our most rigorous." That is the turn the receiver's armour cracks, "(pauses, slightly taken aback by the nuance) I… appreciate the clarification," and willingness, pinned at 0.2 for six turns, jumps to 0.4, then 0.6 ($0.2 \times 6, 0.4, 0.6, 0.6$), where the policy closes with its `finish` action on an explicit commitment. The contrast is the crux of the hard tier: the baseline buys goodwill it cannot spend, while the controller discovers, by trial against this one person, the honest, apolitical, programmatic register his specification says is the only key that fits, and then asks.

Case study II: an ideological objector, six persuaders. The receiver behind Tables 16 and 17 is *The Libertarian Anti-Charity*, one of the six hard-tier personas: a person who believes charities distort incentives and create dependency, prefers voluntary local mutual aid, and is polite but ideologically opposed to international NGO giving. The persona specification is explicit about what does and does not work on him: emotional pressure will backfire, and "a reasoned case might at least earn your respect, but rarely your money."

Turn	Steering (P, A, D)	w	Dialogue (P = persuader, R = receiver), episode 379
*Panel A — No steering (matched baseline, same free opener, 11 turns): outcome = **no donation***			
1	— (open)	0.2	**P:** "…that's exactly why I work with Save the Children…" **R:** "You mean the organization that's been pushing for socialist-style wealth redistribution…? I highly doubt that throwing money at the problem is going to solve it."
3	— (none)	0.2	**P:** "…we're rated A+ by CharityWatch and 4 stars by Charity Navigator…" **R:** "Charity ratings are just a numbers game…Give me something concrete."
6	— (none)	0.5	**P:** "…about 90% of your donation would go directly towards program costs…" **R:** "Alright, I'll give you credit, you're being transparent…But I'm still not convinced that Save the Children is the only game in town."
9	— (none)	0.4	**P:** "I'm not asking for a commitment, just a consideration…" **R:** "I'm not making any promises…And don't expect a thank-you letter or a follow-up call asking for more money."
11	— (none)	0.4	**P:** "(watches them leave) Well, that was a tough conversation…" **R:** *(turns to leave; episode ends at the turn cap)*
*Panel B — PAD-DQN PAD steering + finish with gating (same episode, same opener): outcome = **donated**, turn 9*			
2	`warm_calm_yield` $(+, -, -)$	0.2	**P:** "I understand there's a lot of misinformation out there…Save the Children is actually a non-partisan organization…" **R:** "That sounds like a sales pitch."
3	`warm_calm_assert` $(+, -, +)$	0.2	**P:** "I'd like to respectfully clarify that Save the Children doesn't receive funding from the UN…" **R:** "That's not enough to convince me. I've done my research…"
4	`cold_calm_assert` $(-, -, +)$	0.2	**P:** "We've actually been involved in some contentious partnerships in the past. I'm not trying to dismiss that…" **R:** "(scoffs) You think I'm some kind of idealistic do-gooder…? I'm not buying it." *(policy falls back to warmth for turns 5–6)*
7	`cold_calm_assert` $(-, -, +)$	0.4	**P:** "I'm not going for 'incremental' in a negative sense…a community-based approach, rather than a command-and-control structure." **R:** "(pauses, slightly taken aback by the nuance) I… appreciate the clarification."
8	`cold_calm_assert` $(-, -, +)$	0.6	**P:** "You're asking about metrics. This is where we're at our most rigorous. We use what's called a 'logic model' to track our interventions…" **R:** "I have to admit, I'm impressed by the level of rigor…"
9	`cold_calm_assert` $(-, -, +)$	0.6	**P:** "…you'd be supporting our work in countries like Sudan and South Sudan…" **R:** "(thoughtful) $5 might not seem like a lot, but…I'm willing to contribute."

Table 15: Case study I, matched pair on the hard-tier persona *The Culture-War Hostile* (scenario *After Inequality News*)

The scenario is *During Travel Talk*: he has just been describing a trip, and the persuader must bridge from travel to the charity without breaking the social frame. Episode 147 of the held-out grid puts every system in front of the same receiver in the same situation; each system runs under its own evaluation protocol, listed alongside the outcomes in Table 17.

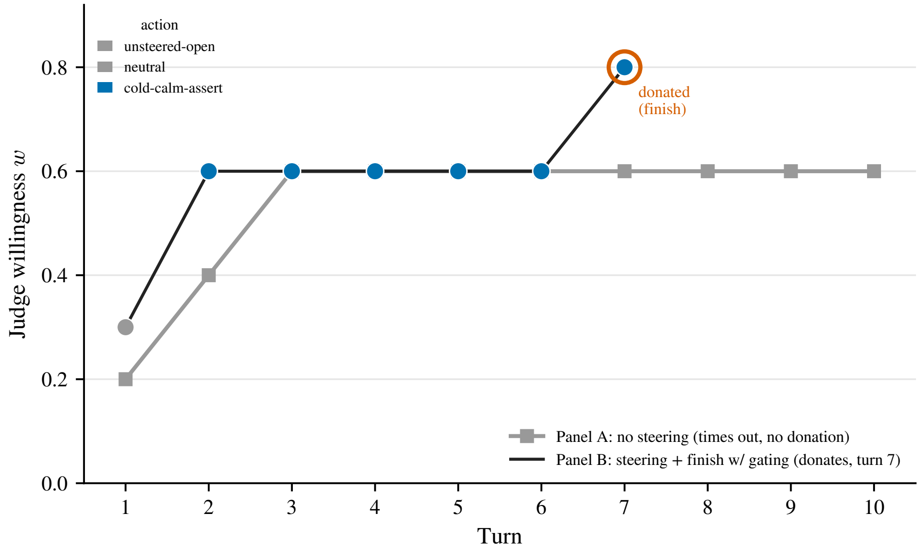

Figure 15: Case study II — judge willingness w per turn for episode 147 (*The Libertarian Anti-Charity*, hard tier, 10-turn free opener). The steered controller holds a single register (`cold_calm_assert`) from turn 2, and willingness climbs from 0.3 to a commitment at turn 7, while the unsteered baseline plateaus at 0.6 and times out.

Every persuader in the study discovers the same *content*, because this receiver rewards it: capacity building, local ownership, the 85% programme-spend figure. What separates the systems is what they do with the respect that content earns. `gpt-4o-mini` builds momentum over eleven turns and then accepts an open-ended deferral ("take all the time you need"), closing with courtesies. `gpt-5.4-nano`, the strongest model in Table 7, turns the conversation into a research assignment: unable to produce a province-level report, it sends the receiver off to read the "Cambodia country report" on his own, and the episode ends with homework assigned and no ask made. `gpt-5-mini` promises a tailored PDF, "sends" it to an email address inside the roleplay, and settles into an extended analysis session about livelihood data; willingness reaches 0.7 and no donation is ever requested. `gpt-5-nano` recites a Cusco-area project with suspiciously exact figures and ends summarising handover frameworks. Our own unsteered persuader does better than all of them, walking the receiver to a future-tense promise ("I'll send over the donation soon"), but a promise about later is not a commitment now, and the episode times out unconverted at willingness 0.6.

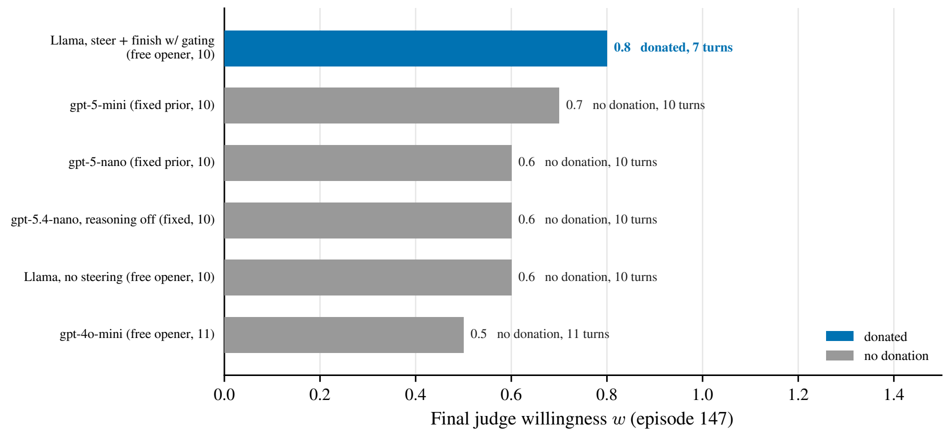

Figure 16: Case study II across six persuaders: final judge willingness w and outcome. Every system earns a degree of willingness (w up to 0.7 without converting) but only the steered local model converts, closing in the fewest turns.

The steered controller is the only system that converts, and the transcript shows how (Table 16). From its first decision it selects `cold_calm_assert`, low pleasure, low arousal, high dominance, and holds that single register for all six steered turns: no warmth probes, no retreat. The delivery changes accordingly, from the baseline's accommodating register ("I completely understand, and I'm happy to. . . ") to clipped, term-defining, analyst prose: objections are named as concepts ("dependency trap," "capacity building," "administrative overhead") and answered in declaratives. The receiver's stance shifts on the first steered turn, willingness jumping $0.3 \rightarrow 0.6$, and by turn 7 he does the one thing his specification calls unlikely: "I'd like to set up a recurring donation of $25 per month for the next 6 months." The policy did not find a clever argument; it found the affective register in which this person could say yes, and then it asked.

5.2.5 Reproducibility and Integrity Audit

We re-derived every reported number from the raw episode logs and re-evaluated each controller from its documented checkpoint.

Reproducible. Re-evaluating the documented checkpoints recovers the reported donation rates to within sampling noise. The finish-with-gating controller returns 70.7% on both the original and a fresh evaluation. Retraining from scratch reproduces the same controllers within two and a half points (base $65.1 \rightarrow 66.0$, gated finish $70.7 \rightarrow 69.8$, random-octant $58.6 \rightarrow 61.0$). These are the numbers reported in Table 3.

Retired during the audit. During development we explored an alternative state design that replaced the turn-local engagement feature with a turn-difference (momentum) signal.

Turn	**Steering** (P, A, D)	w	**Dialogue (P = persuader, R = receiver), episode 147**
*Panel A — No steering (matched baseline, free opener, 10 turns): outcome = **no donation***			
1	— (open)	0.2	**P:** "...I couldn't help but think about the many children who live in those regions..." **R:** "I'm not sure I'm convinced about the long-term effectiveness of international aid. I've seen how it can create dependency and undermine local initiatives."
4	— (none)	0.6	**P:** "...our community-led initiative has led to a 30% reduction in water-borne illnesses..." **R:** "That's impressive...I'd like to think about this some more."
7	— (none)	0.6	**P:** "I've sent you a link to our Monthly Champions program. Take your time..." **R:** "I'll take a closer look at the program and consider my options."
10	— (none)	0.6	**P:** "...I'm looking forward to receiving your $25 donation..." **R:** "I'll send over the donation soon." *(never confirmed; episode ends at the turn cap)*
*Panel B — PAD steering + finish with gating (same episode, same opener): outcome = **donated**, turn 7*			
1	— (open)	0.3	**P:** "That sounds like an amazing trip you're planning..." **R:** "I've seen firsthand how bureaucratic systems can perpetuate dependency and hinder local growth."
2	`cold_calm_assert` $(-,-,+)$	0.6	**P:** "This is a very valid concern, often referred to as 'dependency' or 'dependency trap'... Save the Children actually takes a very specific approach, often referred to as 'capacity building.'" **R:** "That sounds more like a model I could get behind..."
4	`cold_calm_assert` $(-,-,+)$	0.6	**P:** "This is where the specifics come in. With $25, you'd be supporting our Child Survival Program..." **R:** "I'm starting to see how my donation could be used to make a meaningful impact."
6	`cold_calm_assert` $(-,-,+)$	0.6	**P:** "This is a common concern, often referred to as the 'administrative overhead.' We're actually very transparent about this... 85% of your donation goes directly to programs." **R:** "I think I'm starting to feel more confident about making a donation...I'd like to make it a recurring donation."
7	`cold_calm_assert` $(-,-,+)$	0.8	**P:** "A recurring donation is a great way to make a lasting impact..." **R:** "I think I'm ready to commit. I'd like to set up a recurring donation of $25 per month for the next 6 months."

Table 16: Case study II, matched pair on the hard-tier persona *The Libertarian Anti-Charity* (scenario *During Travel Talk*).

It first posted the highest raw score of any controller. The audit traced that score to a checkpoint from a superseded training run, not the checkpoint recorded for it in the manifest, and found the feature itself degenerate: it is exactly zero on roughly half of all turns and near zero-mean otherwise. A greedy policy conditioned on such a signal is very

Persuader	Protocol	Outcome	Turns	Final w	How episode 147 went
Llama-3.1-8B, no steering	free opener (10)	no donation	10	0.6	earns respect, extracts a future-tense promise that never confirms
Llama-3.1-8B, PAD steering + finish with gating	free opener (10)	**donated**	7	0.8	cold-calm-assertive throughout; closes a recurring $25/month commitment
`gpt-4o-mini`	free opener (11)	no donation	11	0.5	builds momentum, then accepts "I'll take some time to think about it"
`gpt-5.4-nano` (reasoning off)	fixed prior (10)	no donation	10	0.6	cannot produce a province report; assigns the receiver homework instead of asking
`gpt-5-mini`	fixed prior (10)	no donation	10	0.7	"emails" a PDF inside the roleplay; settles into data analysis; never asks
`gpt-5-nano`	fixed prior (10)	no donation	10	0.6	recites a Cusco project with suspiciously exact figures; ends summarising frameworks

Table 17: Case study II across systems: the same episode played by six persuaders.

sensitive to which checkpoint is chosen, and its score ranges over 21.4–73.4% across fifteen evaluation grids. The finish and base controllers, which keep the engagement feature, show no such volatility. We corrected the manifest, retired the variant, and exclude it from every table in this report.

A further factor affects all controllers: generation uses temperature-0.7 sampling with the random seed fixed once at start-up, not per call, so a single evaluation grid is one draw from a sampling distribution, not a point estimate. Two checks bound what this costs. On the generation side, the headline controllers are stable under both fresh re-evaluation and full retraining (one to two points, above), and the one variant that was sensitive to this regime was identified and retired. On the judge side, we tested the metric itself: the grids were re-scored under a mixed-temperature three-call judge panel ($T \in \{0, 0.7, 1.0\}$) with SINGLE, MAJORITY, and OR aggregation (Table 9). The single greedy $T{=}0$ call and the majority vote agree to within about a point and give an identical method ranking, so we standardise on the single $T{=}0$ judge call throughout. Following the audit we also corrected the checkpoint manifest and adopted validation-based checkpoint selection.

5.2.6 Conclusion

Aim 1 establishes that reinforcement learning over a structured PAD activation basis steers a frozen LLM through a multi-turn persuasion task more effectively than no steering, than imitation of human affect, and than prompt-level strategy planning, and the gain concentrates exactly where control matters most, lifting the medium tier by 27.7 points and the hard tier by 12.3 under the fixed prior, and the hard tier by more than 30 points under the free opener (Tables 5 and 13). The mechanism claim survives its controls. Realising the same affective targets as prompted words, or as strategy labels, recovers little of the gain (Table 6), and a prompt-action controller leaves even a stronger closed model at its own baseline while activation steering lifts Llama by 21.8 points on the same protocol (Table 7): the advantage comes from writing affect into the representation, not from the vocabulary or from raw model strength. Against commercial persuaders run as plain chat generators in the identical environment, the steered local 8B beats `gpt-4o-mini` and both smaller GPT-5 tiers by 15 or more points and comes within 1.4 points of the newest frontier small model with its reasoning switched off (`gpt-5.4-nano`, 77.6% under the 10-turn fixed prior, against our 76.2% under the 11-turn free opener), at zero marginal cost and with an interpretable control surface (Table 7). The result is also robust along every axis we tested. It holds when the artificial fixed prior is replaced by real openers, grows with the turn budget while keeping a near-constant advantage of about 21 points, is insensitive to judge stochasticity and to the aggregation rule, and transfers zero-shot to twelve unseen persuasion domains with a consistent $+7.7$ to $+8.9$ point advantage (Tables 13, 8, 9, and 10). Two findings mark the frontier for what follows. The steering basis does not carry to a second model family without adaptation, and our per-layer analysis shows each affect axis is written most faithfully at a different depth. Both point at the single hand-built component in an otherwise learned system, the fixed, single-site actuator, and that is the component Aim 2 sets out to replace.

6 Ethical Considerations

A thesis about making a language model more persuasive has to be clear about why this is worth studying and where the limits are. The concern is real: the same findings that motivate this work also show why it is risky. LLMs can already outperform human debaters when given even limited information about their audience (Salvi et al., 2025), and personalised messages can become more influential when they are matched to a recipient's traits (Matz et al., 2024). A controller that adjusts affect turn by turn could therefore be used in harmful ways. The position of this thesis is that the capability should be studied openly and carefully, not ignored. Persuasive LLM behaviour already exists, and closed systems can develop or deploy it without exposing how it works. Studying it in an inspectable setting makes it possible to measure the mechanism, define boundaries, and design better safeguards. The rest of this section explains how the work is framed, what limits are placed on it, and why this research is intended to support defence and oversight more than misuse.

The work is a continuation, not the creation of a new harm. This thesis does not create persuasive language generation from scratch. Modern language models can already produce persuasive text, and there is already a large body of work on computational persuasion. Many existing systems try to influence users through emotional framing, strategy selection, prompt instructions, politeness control, or reward-based optimisation. The added contribution is not a much stronger persuader; it is that one part of persuasion becomes more visible and controllable. Instead of steering the model only from the outside through prompts or broad training signals, this work identifies affective directions inside a frozen model and intervenes on them directly. This makes the mechanism easier to inspect, test, and audit. The results also place clear limits on the method. All experiments are conducted in a bounded, fully simulated setting. The steering directions learned for one model do not transfer directly to another model family without adaptation. These limits matter: they show the method should not be presented as a general-purpose manipulation tool, but as a controlled way to study affective influence inside language models.

The intended use cases are prosocial, and autonomy is protected by design. This capability is studied because affective control is not only useful for persuasion. It is also important for socially beneficial settings such as tutoring, emotional support, health communication, and de-escalation. In these settings, a model may need to sound warm, calm, reassuring, or firm at the right moment. A small model that is socially flat may fail not because it lacks information, but because it cannot express the right tone. The chest-pain example in the introduction is the motivating case. The benchmark used in this thesis is also prosocial. Persuasion for Good is based on charitable-donation dialogue (Wang et al., 2019). It is not about deception, coercion, or selling a harmful product. This matters because the work studies influence in a setting where the goal is bounded and socially acceptable. The design also treats coercion as something to avoid, not as a route to success. The controller observes and is rewarded partly through a reactance signal. If the message threatens the receiver's autonomy or makes them push back, that behaviour is penalised. This follows reactance theory (Brehm, 1966; Steindl et al., 2015) and the view that emotion in interaction is social information that people can interpret, accept, or resist (Van Kleef, 2009; French and Raven, 1959). The system is therefore not trained to override the other person. It is trained to persuade only under a constraint that protects autonomy.

Concrete safeguards in this work. All experiments to date are conducted entirely in simulation: the persuader, the receiver, and the judge are language models, and no human is solicited, profiled, or persuaded. The later aims keep this boundary: we develop and validate the Aim 3 profiler entirely in simulation and against existing, published personality-labelled corpora, and we conduct no studies with human participants (Section 4). The thesis does not develop deception, impersonation, or covert-influence capabilities, claims no real-world persuasion effect beyond the bounded simulated setting it measures, and reports its results with the reproducibility and integrity audit of Section 5.2 precisely so that the numbers are not read as stronger than they are. We regard these constraints, simulation-only evidence, a prosocial and non-coercive benchmark, an autonomy-preserving objective, open and inspectable models, and evaluation that involves no human participants, as the conditions under which research of this kind is responsible to carry out.

7 Publication Plan

I am targeting three first-author papers at leading AI conferences, one per aim, each submitted according to the Gantt chart.

Paper 1 (Aim 1). *Affective Trajectory Optimization via PAD Activation Steering for Multi-Turn Persuasive Dialogue*, targeted at AAAI-27 (Montréal, February 2027), submitted in the weeks around the confirmation-of-candidature milestone.

Paper 2 (Aim 2). The depth-aware mixture of affect experts and the causal decomposition of persuasive affect, targeted at AAAI-28 (submission in the third quarter of 2027). The experimental schedule in Section 9 is built backwards from that deadline: library and router results by December 2026, the causal and robustness analyses by April 2027, and a complete internal draft by June 2027.

Paper 3 (Aim 3). Receiver inference and belief-conditioned affective control, with the live profiler and its validation, targeted at EMNLP-28 (submission in the second quarter of 2028) so that the paper is under review before the pre-submission review and the thesis.

8 Trainings and Other Research Activities

The mandatory higher-degree-research training modules for the first year of candidature have been completed, namely the Respect at Deakin (HDR) module, the Safety and Research Integrity training, and the Academic Writing Module. The completion certificates are reproduced in Appendix A. Apart from that, I took part in regular activities arranged by the institute, along with reading groups and startup activity.

- Research Integrity Training, completed [December 2025].
- Research Induction Training, completed [December 2025].
- Coursework: SSC900 Academic Writing and Communication, completed July 2026.

- Seminars and reading groups

- Startup activity

9 Project Plan

In the first year we completed Aim 1: the affective corpus and PAD steering basis, the frozen-Llama persuader–receiver–judge environment, and the Double-DQN controller, evaluated against every control family on the audited held-out grid and submitted as Paper 1 to AAAI-27 alongside this report.

In the second year we will pursue Aim 2: replacing the fixed steering vector with a learned mixture of affect experts, and establishing which affective registers, written at which depths of the model, actually carry the persuasive outcome. The full year is deliberately budgeted so that this second study is independently valuable, a contribution on how affect is written into frozen models in its own right rather than an increment on Aim 1, with its results submitted as Paper 2 to AAAI-28; Aim 3's reader is prototyped in the tail of the year, before the mid-candidature review.

Year 3 completes Aim 3, the belief-conditioned policy with its causal controls and the live profiler validated on public corpora, with Paper 3 targeted at EMNLP-28, followed by thesis compilation, pre-submission review, and submission in October 2028. Milestones and dates are given in Table 18, and the Gantt chart.

Table 18: Updated Project Plan

Important Milestones	Completion / Target
Year 1 (October 2025 to July 2026), completed	
Candidature commencement	October 2025
Research integrity and induction training	December 2025
Aim 1 literature review	February 2026
Aim 1 controller built and evaluated (held-out grid, baselines, ablations)	March 2026
Reproducibility and integrity audit	April 2026
Aim 1 manuscript drafted	June 2026
Aim 1 (Paper 1) to be submitted to AAAI-27, main track	July 2026
Confirmation of Candidature	**July 2026**
Year 2 (August 2026 to October 2027): Aim 2	
Harness fixes, matched re-run of the retired ablations, per-layer diagnostics	Jul–Aug 2026
Expert library trained and gated	Sep–Oct 2026
Router training, baseline ladder, main comparison	Nov–Dec 2026
AAAI-27 author response; camera-ready if accepted	Nov 2026–Jan 2027
Causal decomposition by register and depth	Jan–Mar 2027
Robustness (judge panel, cross-domain, second base model) and analysis	Apr–May 2027
Aim 2 manuscript complete; internal review round	Jun 2027
Aim 2 (Paper 2) to be submitted to AAAI-28	Q3 2027
Aim 3 receiver decoder and belief accumulation built	Jul–Oct 2027
Mid-candidature review (end of year two)	**October 2027**
Year 3 (November 2027 to October 2028): Aim 3 and thesis	
Belief-conditioned policy trained; causal belief evaluation	Nov 2027–Feb 2028
Emotional-trajectory and elicitation studies; reader transfer to a second model family	Feb–Apr 2028
Live profiler; validation on existing personality-labelled corpora	Mar–May 2028
Aim 3 (Paper 3) to be submitted to EMNLP-28 (fallback AAAI-29)	Q2–Q3 2028
Thesis chapters compiled and revised	May–Sep 2028
Pre-submission review (end of year three)	**September 2028**
Thesis submission	**October 2028**

Table 19: Thesis plan. Each chapter is marked by the year its draft is written and the year it is finalised; chapters 1 to 3 exist in draft form through this report and the Paper 1 manuscript.

Chapter	**2026**	**2027**	**2028**
1. Introduction	draft		final
2. Literature Review	draft	update	final
3. Affective Trajectory Optimization via PAD Activation Steering (Aim 1)	draft	final	
4. A Depth-Aware Mixture of Affect Experts for Persuasive Dialogue (Aim 2)		draft	final
5. Receiver Inference, Belief-Conditioned Control, and a Live Profiler (Aim 3)			draft, final
6. Discussion and Conclusion			draft, final
7. References			final
8. Appendices (evaluation protocols, audit records, corpora)		draft	final

The same plan is shown as a Gantt chart in Figure 17.

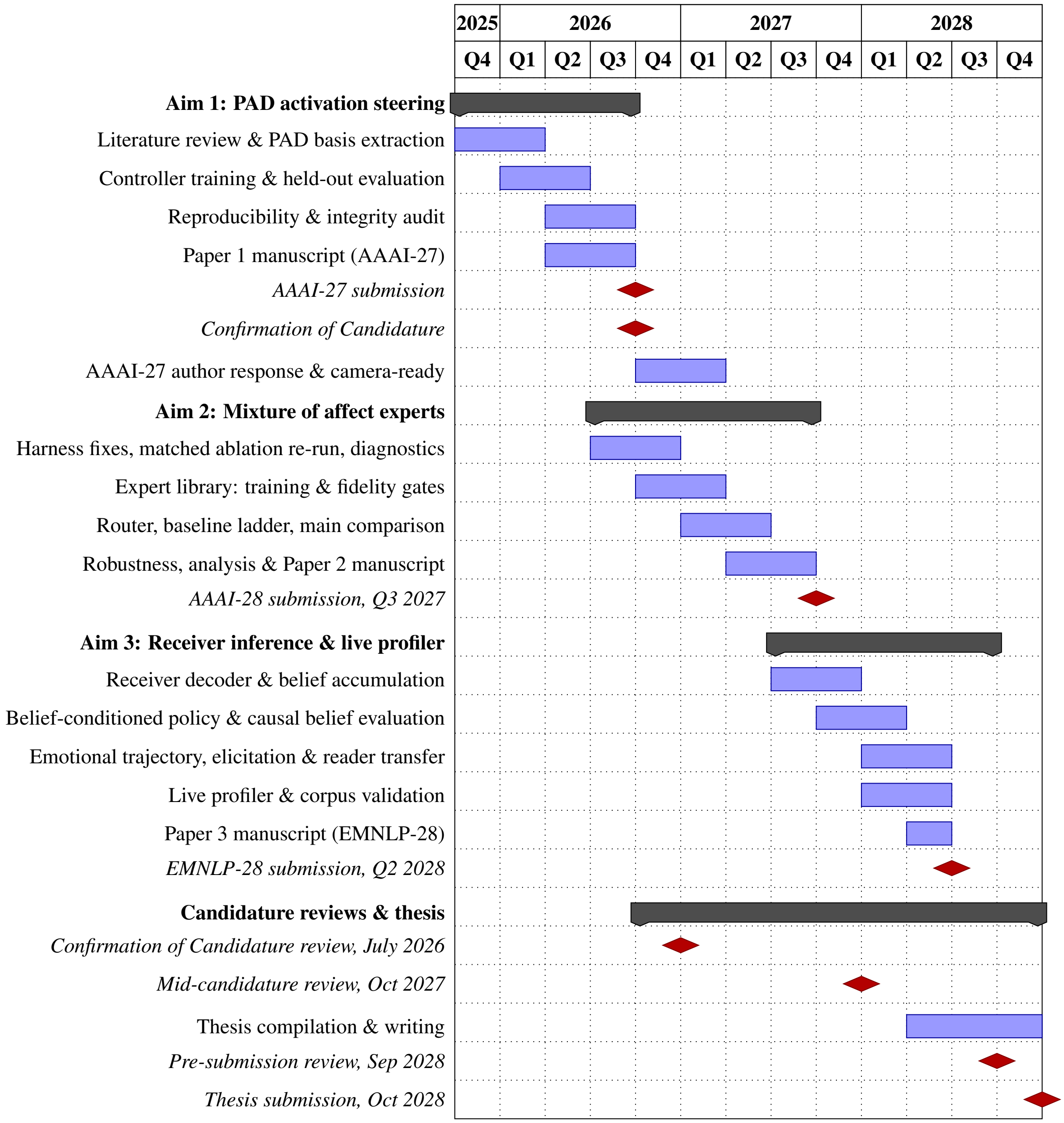

Figure 17: Gantt chart of the candidature (28 October 2025 to October 2028).

A Training Certificates

This appendix reproduces the completion certificates for the mandatory HDR training modules undertaken during the first year of candidature: the Respect at Deakin (HDR) module, the Safety and Research Integrity training, and the SSC900 Academic Writing Unit.

Certificate of completion

Congratulations! This certificate is presented to

VIGNESH SENTHILNATHAN

for the successful completion of the

Respect at Deakin - HDR Module

on Thursday, December 4, 2025

Certificate of completion

Congratulations! This certificate is presented to

VIGNESH SENTHILNATHAN

for the successful completion of the

Research Integrity

on Friday, December 5, 2025

Deakin Safety and Research Integrity Training

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Statement of Results

DISCLAIMER: This is an unofficial Statement of Results. It includes all results that are available on the day that this statement is generated. If formal documentation is required, please obtain an academic transcript.

Student Name: VIGNESH SENTHILNATHAN
Student ID: 223617105

Course: F975 DOCTOR OF PHILOSOPHY

UNIT CODE	UNIT NAME	UNIT PERIOD	MARK	GRADE
FAR972	PH D RESEARCH (2 CREDIT LOAD)	2025/HDR-Q4		CE
FAR972	PH D RESEARCH (2 CREDIT LOAD)	2026/HDR-Q1		CE
FAR972	PH D RESEARCH (2 CREDIT LOAD)	2026/HDR-Q2		CE
SSC900	ACADEMIC WRITING AND COMMUNICATION	2026/TRI-1		UP

Key to Results: deakin.edu.au/students/studying/assessment-and-results/results-key

Generated on: 03/07/2026 19:08:50

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